

Rings That Are
Nearly Associative

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RINGS THAT ARE NEARLY ASSOCIATIVE

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Preface

Until thirty years into this century the theory of rings developed mainly as the theory of associative rings. However, even in the middle of the last century there arose mathematical systems that satisfied all the axioms of a ring except associativity. For example, the algebra of Cayley numbers, which was constructed in 1845 by the English mathematician Arthur Cayley, is such a system. An identity found by Degen in 1818, which represents the product of two sums of eight squares again in the form of a sum of eight squares, was rediscovered by means of this algebra. The algebra of Cayley numbers is an eight-dimensional division algebra over the field of real numbers, which satisfies the following weakened identities of associativity: $(aa)b = a(ab)$, $(ab)b = a(bb)$.

Algebras satisfying these two identities were subsequently named alternative. Such a name stems from the fact that in every algebra satisfying the two indicated identities, the associator $(x, y, z) = (xy)z - x(yz)$ is an alternating (skew-symmetric) function of its arguments. The theory of alternative algebras attracted the serious attention of mathematicians after the discovery of its deep connection with the theory of projective planes, which was actively developed at the beginning of this century. In this regard, it was discovered that alternative algebras are “sufficiently near” to associative ones. The essence of this nearness is exhibited by the theorem of Artin, which asserts that in every alternative algebra the subalgebra generated by any two elements is associative.

Another important class of nonassociative algebras is Jordan algebras, which are defined by the identities $xy = yx$, $(x^2y)x = x^2(yx)$. They arose in the work of the German physicist Jordan, that dealt with the axiomatization of the foundations of quantum mechanics. Jordan algebras are connected with associative algebras in the following manner.

Let A be an associative algebra over a field of characteristic $\neq 2$. We define on the vector space of the algebra A a new operation of multiplication, $a \odot b = \frac{1}{2}(ab + ba)$, and denote the algebra obtained by $A^{(+)}$. This algebra is Jordan. If a subspace J of the algebra A is closed with respect to the operation $a \odot b$, then J together with this operation forms a subalgebra of the algebra $A^{(+)}$, and consequently is a Jordan algebra.

A similar connection exists between associative algebras and Lie algebras, which are defined by the identities $x^2 = 0$, $(xy)z + (zx)y + (yz)x = 0$. If there is defined on an associative algebra A a new multiplication given by the formula $[a, b] = ab - ba$, then the obtained algebra $A^{(-)}$ is a Lie algebra. For Lie algebras over fields there is also a connection in the reverse direction. By the Poincaré–Birkhoff–Witt theorem, every Lie algebra over an arbitrary field is a subalgebra of an algebra $A^{(-)}$ for some suitable associative algebra A .

The situation is different for the theory of Jordan algebras. Over any field there exist Jordan algebras that are not subalgebras of algebras $A^{(+)}$ for any associative algebra A . If Jordan algebra J is a subalgebra of an algebra $A^{(+)}$, where A is an associative algebra, then J is called a special Jordan algebra.

A connection also exists in the reverse direction for Jordan algebras, although it is not as strong as in the case of Lie algebras. The basic result in this direction, namely, that any Jordan algebra with two generators is special, is due to one of the authors.

The theories of alternative, Lie, and Jordan algebras, which are closely interlaced among themselves, have extensive contacts with different areas of mathematics, and together with the highly developed theory of associative rings comprise the basic nucleus of the modern theory of rings. Algebras from these three classes are combined under the general title “algebras that are nearly associative.” This term was introduced in a survey article by A. I. Shirshov [279], where an extensive program for further investigations was outlined.

Naturally, we could not include the whole theory of rings in our book. The title of the book itself indicates that the theory of associative rings is excluded. However, for reading of the basic text, only a knowledge of such classical material as the structure of Artinian rings or the theory of the Jacobson radical is required of the reader, and this can be read in any book on the theory of associative rings. In addition, our book completely avoids Lie algebras, to which several books in the Russian language are already devoted.

We present in the most detail the theory of alternative algebras, with which the reader previously could become acquainted only through journals. There exist some omissions in our presentation of the theory of Jordan algebras. Many of the results not included are contained in the monograph of Jacobson [47], with which we have tried not to intersect. We have made our accent on the achievements of the last decade.

The first chapter of the book contains reference material. In it are found collections of basic facts on varieties of algebras and linearizations of identities. It is recommended that the reader turn to it as needed. It is also not necessary to read the remaining chapters in succession. We recommend two courses to the reader:

“alternative algebras”—2, 3.1, 3.2, 5–13;

“Jordan algebras”—2–4, 5.1–5.4, 8.1, 8.2, 14, 15;

(indicating only the numbers of chapters and sections). The last chapter, which is devoted to right alternative algebras, contains a basic survey.

Almost every section of the book is provided with exercises, in which it is often suggested that the reader prove a theorem of entirely independent value. In these cases detailed hints are given. The exercises are not used anywhere in the basic text.

In each chapter the numbering of theorems, lemmas, and formulas is its own. In references to theorems (formulas) of another chapter, the number of the appropriate chapter is written before the number of the theorem (formula).

A list of literature, which is not claimed to be complete, is found at the end of the book. The surveys and monographs cited in that list are starred.

Throughout the book there is material used in special courses, which was repeatedly presented by A. I. Shirshov at MGU and NGU,[†] and also used in special courses presented in various years at NGU by the other authors.

The authors thank Yu. A. Medvedev and V. G. Skosyrskiy for their great assistance in putting together the bibliography.

[†] MGU and NGU are the transliterated initials for Moscow State University and Novosibirsk State University, respectively (Translator).

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CHAPTER 1

Varieties of Algebras

1. FREE ALGEBRAS

Let Φ be an associative and commutative ring with identity element 1. A set A is called an *algebra* over the ring Φ if there is defined on A the structure of a unital Φ -module and an operation of multiplication connected with the module operations by the relations

$$(a + b)c = ac + bc, \quad a(b + c) = ab + ac, \\ \alpha(ab) = (\alpha a)b = a(\alpha b),$$

for $a, b, c \in A$ and $\alpha \in \Phi$. We shall call algebras over a ring Φ simply Φ -algebras. Sometimes they are also called Φ -operator rings. The ring Φ is called a *ring of scalars* or *ring of operators*. If Φ is a field, then every customary algebra over the field Φ is obviously a Φ -algebra. Every ring is an algebra over the ring of integers $\mathbb{Z}$. Thus the concept of an algebra over a ring combines the concepts of a ring and an algebra over a field.

Let A and B be two Φ -algebras. A mapping $\phi: A \rightarrow B$ which is a homomorphism of the Φ -modules of these algebras with

$$\phi(ab) = \phi(a)\phi(b) \quad \text{for } a, b \in A$$

is called a *homomorphism* of the Φ -algebra A into the Φ -algebra B . Homomorphisms of Φ -algebras correspond to Φ -admissible ideals, i.e., ideals invariant under multiplication by elements from Φ .

We fix an arbitrary set $X = \{x_\alpha\}$ and add to it two more symbols, left and right parentheses, to obtain a set $X^* = X \cup \{(\,,\,)\}$. Consider all possible finite sequences of elements from the set X^* . Two sequences $a_1 a_2 \dots a_m$ and $b_1 b_2 \dots b_n$, where $a_i, b_j \in X^*$, are considered equal if $m = n$ and $a_i = b_i$ for $i = 1, 2, \dots, m$. We define inductively a set $V[X]$ of those sequences of elements from the set X^* which we shall call *nonassociative words* from elements of the set X . First, all elements of the set X belong to the set $V[X]$. Second, if $x_1, x_2 \in X$ and $u, v \in V[X] \setminus X$, then the sequences $x_1 x_2$, $x_1(u)$, $(v)x_2$, and $(u)(v)$ also belong to the set $V[X]$. No other sequences are contained in $V[X]$. For example, the sequence $(x_1(x_2 x_3))x_4$ is a nonassociative word from elements of the set $X = \{x_1, x_2, x_3, x_4\}$, but the sequence $(x_1(x_2 x_3)x_4)$ is not. The number of elements from the set X which appear in a word v is called the *length of the nonassociative word* v , and it is denoted by $d(v)$.

PROPOSITION 1. Let v be a nonassociative word from elements of some set. Then

- (1) the number of left parentheses in v equals the number of right parentheses;
- (2) in any initial subsequence of the word v , the number of left parentheses is not less than the number of right parentheses.

The proof is carried out by an obvious induction on the length of the word v .

We define on the set $V[X]$ a binary operation, denoted by $\cdot$, according to the following rules. Let $x_1, x_2 \in X$ and $u, v \in V[X] \setminus X$. We set

$$\begin{aligned} x_1 \cdot x_2 &= x_1 x_2, & x_1 \cdot u &= x_1(u), \\ v \cdot x_2 &= (v)x_2, & u \cdot v &= (u)(v). \end{aligned}$$

PROPOSITION 2. Every nonassociative word v with $d(v) \geq 2$ has a unique representation in the form of a product of two nonassociative words of lesser length.

PROOF. The existence of such a representation follows directly from the definition. Suppose that the word v has two different representations, $v = u \cdot w = u' \cdot w'$. Consider the case when the words u, w, u', w' all have length greater than 1. Then

$$v = (u)(w) = (u')(w'),$$

and also $d(u) + d(w) = d(u') + d(w')$. If $d(u) = d(u')$, then it is obvious $u = u'$, $w = w'$. If $d(u) > d(u')$, then the subsequence u' is an initial subsequence of the word u , which by the first part of Proposition 1 has more right parentheses than left. But this contradicts the second part of that same proposition. The analysis of the remaining cases is even simpler. This proves the proposition.

We now consider the free unital Φ -module $\Phi[X]$ from the set of generators $V[X]$, and we extend the operation of multiplication defined on $V[X]$ to $\Phi[X]$ by the rule

$$\left(\sum_i \alpha_i u_i\right) \cdot \left(\sum_j \beta_j v_j\right) = \sum_{i,j} \alpha_i \beta_j (u_i \cdot v_j),$$

where $\alpha_i, \beta_j \in \Phi$ and $u_i, v_j \in V[X]$. We have obtained a Φ -algebra $\Phi[X]$ which is called the *free algebra* over the ring Φ from set of generators X . Free algebras possess the following universal property.

THEOREM 1. Let A be an arbitrary Φ -algebra and θ be some mapping of X into A . Then θ extends uniquely to a homomorphism of the algebra $\Phi[X]$ into A .

PROOF. We first extend θ to $V[X]$. Suppose that θ is already defined on the set of all nonassociative words of length less than n , and let the length of the word w equal n . By Proposition 2, w is uniquely representable in the form of a product of two words of lesser length, $w = u \cdot v$. The elements $\theta(u)$ and $\theta(v)$ are already defined, so therefore we set $\theta(w) = \theta(u) \cdot \theta(v)$. This mapping is well defined since u and v are uniquely determined by the word w . We now extend θ to $\Phi[X]$ by setting

$$\theta\left(\sum_i \alpha_i u_i\right) = \sum_i \alpha_i \theta(u_i)$$

for all $\alpha_i \in \Phi$ and $u_i \in V[X]$. This can be done since $\Phi[X]$ is a free Φ -module. It is not difficult now to see that the indicated mapping θ is a homomorphism of $\Phi[X]$ into A . Its uniqueness is clear. This proves the theorem.

The elements of the algebra $\Phi[X]$ are called *nonassociative polynomials* from elements of the set X . A nonassociative polynomial of the form αv , where $\alpha \in \Phi$ and $v \in V[X]$, is called a *nonassociative monomial*. The length of the word v is called the *degree of the monomial*. The maximum of the degrees of the monomials whose sum forms a polynomial is called the *degree of the polynomial*.

2. VARIETIES. ALGEBRAS FREE IN A VARIETY

We fix a countable set of symbols $X = \{x_1, x_2, \dots\}$. Let f be an arbitrary element from $\Phi[X]$. There appear in its list only a finite number of elements from X , for example $x_1, x_2, \dots, x_n$. In this case we shall write $f = f(x_1, x_2, \dots, x_n)$.

Let A be some Φ -algebra and $a_1, a_2, \dots, a_n \in A$. By Theorem 1 there exists a unique homomorphism $\theta: \Phi[X] \rightarrow A$ which sends x_i to a_i for $i = 1,$

$2, \dots, n$, and sends the other elements of the set X to zero. We shall denote the image of the element f under the homomorphism θ by $f(a_1, a_2, \dots, a_n)$, and say that the element $f(a_1, a_2, \dots, a_n)$ is obtained by substitution of the elements $a_1, a_2, \dots, a_n$ in the nonassociative polynomial $f(x_1, x_2, \dots, x_n)$.

A nonassociative polynomial $f = f(x_1, x_2, \dots, x_n) \in \Phi[X]$ is called an *identity* of the algebra A if $f(a_1, a_2, \dots, a_n) = 0$ for any $a_1, a_2, \dots, a_n \in A$. We also say that A satisfies the identity f or that the identity f is valid in A . The collection of all identities for a given algebra is an ideal of the algebra $\Phi[X]$, which is called the *identities ideal* (*T-ideal*) of the algebra A and is denoted $T(A)$. The collection of all identities which are satisfied by each algebra from some class of algebras $\mathfrak{M}$ is also an ideal in $\Phi[X]$. It is called the *identities ideal* (*T-ideal*) of the class of algebras $\mathfrak{M}$ and is denoted $T(\mathfrak{M})$.

Let I be a subset in $\Phi[X]$. The class $\mathfrak{M}$ of all Φ -algebras satisfying each identity from I is called the *variety* of Φ -algebras defined by the set of identities I . I is called a set of *defining identities* for the variety $\mathfrak{M}$. For example, all associative Φ -algebras form a variety with the one defining identity $f = (x_1 x_2) x_3 - x_1 (x_2 x_3)$. The class of all Lie Φ -algebras is also a variety, whose defining identities are $f_1 = x_1^2$, $f_2 = (x_1 x_2) x_3 + (x_2 x_3) x_1 + (x_3 x_1) x_2$. The identities $f_1 = x_1^2 x_2 - x_1 (x_1 x_2)$ and $f_2 = x_1 x_2^2 - (x_1 x_2) x_2$ define the variety of alternative algebras. Individually these identities define the varieties of left alternative and right alternative algebras, respectively. The variety of Jordan algebras is defined by the identities $f_1 = x_1 x_2 - x_2 x_1$, $f_2 = (x_1^2 x_2) x_1 - x_1^2 (x_2 x_1)$.

The variety consisting of only the zero algebra is called *trivial*.

Let $\mathfrak{M}$ be a nontrivial variety of Φ -algebras and F be an algebra from $\mathfrak{M}$ with set of generators Y . The algebra F is called *free in the variety* $\mathfrak{M}$ (or $\mathfrak{M}$ -free) with set of free generators Y if every mapping of the set Y into an arbitrary algebra A from $\mathfrak{M}$ can be extended uniquely to a homomorphism of the algebra F into A .

Let I be some set of polynomials from $\Phi[X]$. We shall denote by $I(A)$ the ideal of the Φ -algebra A generated by all possible elements of the form $f(a_1, a_2, \dots, a_n)$, where $f \in I$ and $a_1, a_2, \dots, a_n \in A$.

THEOREM 2. Let $\mathfrak{M}$ be a nontrivial variety of Φ -algebras with system of defining identities I . Then for any set Y the restriction to Y of the canonical homomorphism $\sigma: \Phi[Y] \rightarrow \Phi[Y]/I(\Phi[Y])$ is injective and the algebra $\Phi_{\mathfrak{M}}[Y^\sigma] = \Phi[Y]/I(\Phi[Y])$ is free in the variety $\mathfrak{M}$ with set of free generators Y^σ . Any two free algebras in $\mathfrak{M}$ with equivalent sets of free generators are isomorphic.

PROOF. The algebra $\Phi_{\mathfrak{M}}[Y^\sigma]$ is generated by the set Y^σ and belongs to $\mathfrak{M}$ since it satisfies all identities from I . We shall show that it is free in $\mathfrak{M}$ with set of free generators Y^σ . Let $A \in \mathfrak{M}$ and $\tau: Y^\sigma \rightarrow A$ be some mapping of

Y^σ into A . Then by Theorem 1 the mapping $\sigma \circ \tau: Y \rightarrow A$ extends to a homomorphism $\eta: \Phi[Y] \rightarrow A$.[†] But the ideal $I(\Phi[Y])$ is contained in the kernel of any homomorphism of the algebra $\Phi[Y]$ into an algebra of the variety $\mathfrak{M}$, and so in particular it is in the kernel of the homomorphism η . Therefore there exists a homomorphism $\phi: \Phi[Y]/I(\Phi[Y]) \rightarrow A$ which makes the following diagram commutative:

$$\begin{array}{ccc} \Phi[Y] & \xrightarrow{\sigma} & \Phi[Y]/I(\Phi[Y]) \\ & \searrow \eta & \downarrow \phi \\ & & A \end{array}$$

that is, such that $\sigma \circ \phi = \eta$. If $y \in Y$, then $\phi(y^\sigma) = y^\sigma \circ \phi = y^\eta = y^\sigma \circ \tau = \tau(y^\sigma)$, and this means ϕ extends the mapping τ . Consequently ϕ is the desired homomorphism. Its uniqueness is clear.

Thus $\Phi_{\mathfrak{M}}[Y^\sigma]$ is a free algebra in the variety $\mathfrak{M}$, and Y^σ is the set of its free generators. We shall prove that the restriction of σ to Y is injective. Let $y_1 \neq y_2$ but $y_1^\sigma = y_2^\sigma$. We consider a nonzero algebra A from $\mathfrak{M}$ and a nonzero element a from A . There exists a homomorphism $\tau: \Phi[Y] \rightarrow A$ such that $y_1^\tau = a$, $y_2^\tau = 0$. Since $I(\Phi[Y])$ is contained in the kernel of τ , there exists a homomorphism $\phi: \Phi_{\mathfrak{M}}[Y^\sigma] \rightarrow A$ such that $\tau = \sigma \circ \phi$. But now, since $y_1^\sigma = y_2^\sigma$, we have $a = y_1^\tau = (y_1^\sigma)^\phi = (y_2^\sigma)^\phi = y_2^\tau = 0$. This contradiction proves the injectivity of the restriction of σ to Y .

We have proved that Y^σ is a set of free generators for the algebra $\Phi_{\mathfrak{M}}[Y^\sigma]$ which is equivalent to Y . By means of a standard procedure we can now construct a free algebra $\Phi_{\mathfrak{M}}[Y]$ with set of free generators Y .

Now suppose the sets Y and Y' are equivalent. We shall prove that the algebras $\Phi_{\mathfrak{M}}[Y]$ and $\Phi_{\mathfrak{M}}[Y']$ are isomorphic. Consider a bijection $\sigma: Y \rightarrow Y'$. By what has already been proved, there exists a unique homomorphism $\tau: \Phi_{\mathfrak{M}}[Y] \rightarrow \Phi_{\mathfrak{M}}[Y']$ extending σ . Consider also the bijection $\sigma^{-1}: Y' \rightarrow Y$ and the homomorphism $\tau': \Phi_{\mathfrak{M}}[Y'] \rightarrow \Phi_{\mathfrak{M}}[Y]$ extending σ^{-1} . Then $\tau \circ \tau'$ homomorphically maps the algebra $\Phi_{\mathfrak{M}}[Y]$ to itself, and the restriction of $\tau \circ \tau'$ to Y is the identity mapping. By what has been proved, however, there is only one homomorphism of the algebra $\Phi_{\mathfrak{M}}[Y]$ into $\Phi_{\mathfrak{M}}[Y]$ that is the identity on Y . That homomorphism is the identity mapping, which means $\tau \circ \tau' = \text{id}$. Analogously we have $\tau' \circ \tau = \text{id}$. Consequently τ is an isomorphism.

This proves the theorem.

As is easy to note, the varieties of associative, alternative, right alternative, and Jordan algebras are nontrivial (e.g., they all contain the algebras with zero multiplication). Therefore they possess free algebras from any set of generators Y , which we shall denote by $\text{Ass}[Y]$, $\text{Alt}[Y]$, $\text{RA}[Y]$, and $\text{J}[Y]$, respectively.

[†] Here and subsequently $(\phi \circ \psi)(x) = \psi(\phi(x))$.

COROLLARY. If I is a system of defining identities for the variety $\mathfrak{M}$, then $T(\mathfrak{M}) = I(\Phi[X])$.

PROOF. The ideal $T(\mathfrak{M})$ is contained in the kernel of any homomorphism of the algebra $\Phi[X]$ onto an algebra from $\mathfrak{M}$. In particular, it is in the kernel of the canonical homomorphism $\sigma: \Phi[X] \rightarrow \Phi_{\mathfrak{M}}[X]$, which equals $I(\Phi[X])$. Since the reverse containment is clear, this proves the corollary.

REMARK 1. In the study of algebras from a specific variety $\mathfrak{M}$ we shall often call an element $f \in \Phi_{\mathfrak{M}}[X]$ an identity of an algebra A from $\mathfrak{M}$, meaning by this that some inverse image (and consequently all inverse images) in $\Phi[X]$ of the element f under the canonical homomorphism of $\Phi[X]$ onto $\Phi_{\mathfrak{M}}[X]$ is an identity of the algebra A .

3. HOMOGENEOUS IDENTITIES AND HOMOGENEOUS VARIETIES

Every nonassociative polynomial $f \in \Phi[X]$ can be uniquely decomposed into an irreducible sum of monomials. We shall say that a monomial αv , where $\alpha \in \Phi$ and $v \in V[X]$, has type $[n_1, n_2, \dots, n_k]$ if the nonassociative word v contains x_i exactly n_i times, and also $n_k \neq 0$ but $n_j = 0$ for $j > k$. For example, the monomial $((x_1 x_2) x_2)(x_1 x_4)$ has type $[2, 2, 0, 1]$. We shall call the number n_i the *degree of the monomial* αv in x_i . If all the monomials in an irreducible listing of the polynomial f have one and the same degree n_i in x_i , then we call the polynomial f *homogeneous in* x_i of degree n_i . Monomials having one and the same type are said to be of the *same-type*. A nonassociative polynomial (identity of an algebra) is called *homogeneous* if all its monomials in an irreducible listing are of the same-type. In other words, a polynomial f is homogeneous if it is homogeneous in each variable. For example, the polynomial $(x_1^2 x_2) x_4 + ((x_1 x_4) x_2) x_1 - x_1^2 (x_2 x_4) - (x_1 x_4)(x_2 x_1)$ is homogeneous since each of its monomials has type $[2, 1, 0, 1]$. The polynomial $x_1^2 x_2 - (x_1 x_3) x_1 + x_1^2$ is homogeneous in x_1 of degree 2; however, it is not homogeneous. A homogeneous polynomial of type $[n_1, n_2, \dots, n_k]$, where all $n_i \leq 1$, is called *multilinear*.

Since the free algebra $\Phi[X]$ is a free Φ -module with set $V[X]$ of free generators, it has a decomposition into a direct sum of submodules $\Phi^{[n_1, \dots, n_k]}[X]$ consisting of the homogeneous polynomials of type $[n_1, \dots, n_k]$

$$\Phi[X] = \bigoplus \Phi^{[n_1, \dots, n_k]}[X].$$

The *degree of a nonassociative polynomial in* x_i is defined as the maximum degree in x_i of its monomials.

Let $f \in \Phi[X]$ be an arbitrary nonassociative polynomial. We group together the same type monomials of the polynomial f . Then f is represented

in the form of a sum of homogeneous polynomials. These homogeneous polynomials are called the *homogeneous components* of the polynomial f .

THEOREM 3. Let $f \in \Phi[X]$ be an identity of a Φ -algebra A of degree k_i in x_i . Set $k = \max k_i$. Suppose that in Φ there are $k + 1$ elements $\alpha_1, \alpha_2, \dots, \alpha_{k+1}$ such that for the Vandermonde determinant

$$d = \begin{vmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^k \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^k \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_{k+1} & \alpha_{k+1}^2 & \cdots & \alpha_{k+1}^k \end{vmatrix} = \prod_{1 \leq i < j \leq k+1} (\alpha_j - \alpha_i)$$

the equation $d \cdot a = 0$ implies $a = 0$ for any $a \in A$. Then every homogeneous component of the identity f is an identity of the algebra A .

PROOF. First we represent f in the form $f = f_0 + f_1 + \cdots + f_{k_1}$, where f_i is the sum of all monomials of the polynomial f having degree i in x_1 . Let $f = f(x_1, x_2, \dots, x_n)$ and $a_1, a_2, \dots, a_n$ be arbitrary elements of the algebra A . For brevity we shall denote $f_i(a_1, a_2, \dots, a_n)$ by $f_i(a)$. For $i = 1, 2, \dots, k_1 + 1$ we have the relation

$$f_0(a) + \alpha_i f_1(a) + \alpha_i^2 f_2(a) + \cdots + \alpha_i^{k_1} f_{k_1}(a) = f(\alpha_i a_1, a_2, \dots, a_n) = 0,$$

whence it follows (see S. Lang, "Algebra," p. 376, Mir, 1968) that $d_1 f_j(a) = 0$, $j = 0, 1, \dots, k_1$, where

$$d_1 = \begin{vmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{k_1} \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{k_1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_{k_1+1} & \alpha_{k_1+1}^2 & \cdots & \alpha_{k_1+1}^{k_1} \end{vmatrix}.$$

Since d_1 is a divisor of d , from $d_1 f_j(a) = 0$ it follows that $f_j(a) = 0$. Since $a_1, a_2, \dots, a_n$ were arbitrary, this means that $f_0, f_1, \dots, f_{k_1}$ are identities of the algebra A . Performing the same operation with the polynomials $f_j, j = 0, 1, 2, \dots, k_1$, and also with the variable x_2 etc., we finally prove the theorem.

The conditions of the theorem are automatically satisfied in each of the following cases:

- (1) there exist $\alpha_1, \alpha_2, \dots, \alpha_{k+1} \in \Phi$ such that the Vandermonde determinant $d = \prod_{i < j} (\alpha_j - \alpha_i)$ is invertible in Φ ;
- (2) the algebra A is a free Φ -module and there exist elements $\alpha_1, \alpha_2, \dots, \alpha_{k+1} \in \Phi$ such that $d = \prod_{i < j} (\alpha_j - \alpha_i) \neq 0$;
- (3) Φ is a field containing more than k elements.

COROLLARY 1. Let f be an identity of the Φ -algebra A of degree k_i in x_i and $k = \max k_i$. Then in any of the cases (1)–(3) every homogeneous component of f is an identity of the algebra A .

A variety $\mathfrak{M}$ is called *homogeneous* if for every $f \in T(\mathfrak{M})$ all homogeneous components of f also belong to $T(\mathfrak{M})$.

COROLLARY 2. Every variety of algebras over an infinite field is homogeneous.

We note that for a homogeneous variety $\mathfrak{M}$

$$T(\mathfrak{M}) = \bigoplus_{[n_1, \dots, n_k]} (\Phi^{[n_1, \dots, n_k]}[X] \cap T(\mathfrak{M})).$$

Therefore the decomposition of the algebra $\Phi[X]$ into a direct sum of the subspaces $\Phi^{[n_1, \dots, n_k]}[X]$ also induces an analogous decomposition for the free algebra $\Phi_{\mathfrak{M}}[X]$ of a variety $\mathfrak{M}$, which is isomorphic to the quotient algebra $\Phi[X]/T(\mathfrak{M})$:

$$\Phi_{\mathfrak{M}}[X] = \bigoplus_{[n_1, \dots, n_k]} \Phi_{\mathfrak{M}}^{[n_1, \dots, n_k]}[X],$$

where

$$\Phi_{\mathfrak{M}}^{[n_1, \dots, n_k]}[X] = \Phi^{[n_1, \dots, n_k]}[X] / \Phi^{[n_1, \dots, n_k]}[X] \cap T(\mathfrak{M}).$$

Therefore for elements of the free algebra $\Phi_{\mathfrak{M}}[X]$ of a variety $\mathfrak{M}$ the concepts of homogeneity, degree, and degree in each of the variables can be reasonably defined.

Later on we shall need the following criterion for homogeneity of a variety.

LEMMA 1. For the homogeneity of a variety $\mathfrak{M}$ it is necessary and sufficient that $T(\mathfrak{M})$ as an ideal of the algebra $\Phi[X]$ has a system of homogeneous generating elements.

PROOF. If $\mathfrak{M}$ is homogeneous, then the collection of all homogeneous components of elements of $T(\mathfrak{M})$ is contained in $T(\mathfrak{M})$ and generates this ideal even as a Φ -module. If $T(\mathfrak{M})$ is generated as an ideal by a set of homogeneous polynomials $\{f_\alpha\}$, then $T(\mathfrak{M})$ is generated as a Φ -module by the homogeneous elements of the form

$$v(u_1, \dots, u_k, f_\alpha, u_{k+1}, \dots, u_m),$$

where $v, u_1, \dots, u_m$ are monomials from $\Phi[X]$. The presence in $T(\mathfrak{M})$ of a system of homogeneous generators as a Φ -module implies the homogeneity of the variety $\mathfrak{M}$. This proves the lemma.

4. PARTIAL LINEARIZATIONS OF IDENTITIES

We fix some element $y \in \Phi[X]$ and define a family $\Delta(y) = \{\Delta_i^k(y) \mid i = 1, 2, \dots; k = 0, 1, 2, \dots\}$ of linear mappings of the algebra $\Phi[X]$ into itself, letting

- (1) $\Delta_i^0(y) = \text{id}$ —the identity mapping;
- (2) $x_s \Delta_i^k(y) = 0$ if $k > 1$ or $k = 1, i \neq s$;
- (3) $x_i \Delta_i^1(y) = y$;
- (4) $(u \cdot v) \Delta_i^k(y) = \sum_{r+s=k} (u \Delta_i^r(y)) \cdot (v \Delta_i^s(y))$;

where $x_j \in X$ and u, v are arbitrary monomials from $\Phi[X]$. As is easy to see, conditions (1)–(4) together with the condition of linearity uniquely define the action of the mappings $\Delta_i^k(y)$ on the algebra $\Phi[X]$.

LEMMA 2. Let $v = v(x_1, \dots, x_n)$ be a multilinear monomial. Then

$$v(f_1, \dots, f_n) \Delta_i^k(y) = \sum_{k_1 + \dots + k_n = k} v(f_1 \Delta_i^{k_1}(y), \dots, f_n \Delta_i^{k_n}(y))$$

for any $f_1, f_2, \dots, f_n \in \Phi[X]$.

PROOF. For brevity we shall denote $\Delta_i^k(y)$ simply by Δ_i^k and $v(f_1, \dots, f_n)$ by $v(f)$. We carry out an induction on n . For $n = 1$ everything is clear. Now let $n > 1$, so that $v = v_1 \cdot v_2$ where $d(v_1) = m < n$, $d(v_2) = l < n$. From the definition of the operator Δ_i^k and the induction assumption, we obtain

$$\begin{aligned} v(f) \Delta_i^k &= (v_1(f) \cdot v_2(f)) \Delta_i^k = \sum_{r+s=k} (v_1(f) \Delta_i^r) (v_2(f) \Delta_i^s) \\ &= \sum_{r+s=k} \left(\sum_{r_1 + \dots + r_m = r} v_1(f_1 \Delta_i^{r_1}, \dots, f_m \Delta_i^{r_m}) \right) \\ &\quad \times \left(\sum_{s_1 + \dots + s_l = s} v_2(f_{m+1} \Delta_i^{s_1}, \dots, f_n \Delta_i^{s_l}) \right) \\ &= \sum_{k_1 + \dots + k_n = k} v_1(f_1 \Delta_i^{k_1}, \dots, f_m \Delta_i^{k_m}) v_2(f_{m+1} \Delta_i^{k_{m+1}}, \dots, f_n \Delta_i^{k_n}) \\ &= \sum_{k_1 + \dots + k_n = k} v(f_1 \Delta_i^{k_1}, \dots, f_n \Delta_i^{k_n}). \end{aligned}$$

This proves the lemma.

COROLLARY. Let $u = u(x_1, \dots, x_n)$ be an arbitrary monomial from $\Phi[X]$ of degree m in x_i . Then for $k \leq m$

$$u \Delta_i^k(y) = u_1 + u_2 + \dots + u_{\binom{m}{k}},$$

where

$$u_1, u_2, \dots, u_{\binom{m}{k}},$$

are all the possible monomials obtainable from u by substituting y for k occurrences of x_i , and for $k > m$

$$u \Delta_i^k(y) = 0.$$

For the proof, it suffices to consider the multilinear monomial

$$v(x_{11}, \dots, x_{1k_1}, \dots, x_{n1}, \dots, x_{nk_n}),$$

for which $u(x_1, \dots, x_n) = v(x_1, \dots, x_1, \dots, x_n, \dots, x_n)$, and to use Lemma 2 as well as condition (2) in the definition of the operator $\Delta_i^k(y)$.

We list some examples:

$$\begin{aligned} [(x_1^2 x_2) x_1] \Delta_1^1(y) &= (x_1^2 x_2) y + [(x_1 y) x_2] x_1 + [(y x_1) x_2] x_1, \\ [(x_1^2 x_2) x_1] \Delta_1^2(y) &= [(x_1 y) x_2] y + [(y x_1) x_2] y + (y^2 x_2) x_1, \\ [(x_1^2 x_2) x_1] \Delta_1^3(y) &= (y^2 x_2) y, \\ [(x_1^2 x_2) x_1] \Delta_1^4(y) &= 0. \end{aligned}$$

Note that the symbol i in the notation $\Delta_i^k(y)$ is needed only as an indication of the generator x_i on which the operator $\Delta_i^k(y)$ acts in a nontrivial fashion. If the elements of the set X are not equipped with indices, we shall write the generating element itself in place of the index. For example,

$$[(x^2 z) t] \Delta_x^1(y) = [(xy) z] t + [(yx) z] t.$$

LEMMA 3. Let $f = f(x_1, \dots, x_n)$ be a polynomial from $\Phi[X]$ of degree m in x_i . Then

$$f(x_1, \dots, x_i + y, \dots, x_n) = \sum_{k=0}^m f \Delta_i^k(y).$$

PROOF. In view of the corollary to Lemma 2 and the linearity of the operator $\Delta_i^k(y)$, it suffices to consider the case when f is a monomial. We carry out an induction on the degree of f . If $d(f) = 1$, then everything is clear. Now let $d(f) > 1$, so that $f = f_1 \cdot f_2$ where $d(f_i) < d(f)$, and let f_1 have degree m_1 in x_i , f_2 have degree m_2 in x_i . By the induction assumption and by the definition of the operator $\Delta_i^k(y)$, we obtain

$$\begin{aligned} f(x_1, \dots, x_i + y, \dots, x_n) &= f_1(x_1, \dots, x_i + y, \dots, x_n) f_2(x_1, \dots, x_i + y, \dots, x_n) \\ &= \left[\sum_{k=0}^{m_1} f_1 \Delta_i^k(y) \right] \left[\sum_{r=0}^{m_2} f_2 \Delta_i^r(y) \right] \\ &= \sum_{j=0}^m \left[\sum_{k+r=j} (f_1 \Delta_i^k(y)) (f_2 \Delta_i^r(y)) \right] = \sum_{j=0}^m f \Delta_i^j(y). \end{aligned}$$

This proves the lemma.

Consider some nonassociative polynomial $f = f(x_1, \dots, x_n) \in \Phi[X]$, and let $x_j \in X \setminus \{x_1, \dots, x_n\}$. We shall call the polynomials $f_i^{(k)}(x_1, \dots, x_n; x_j) = f \Delta_i^k(x_j)$ the *partial linearizations of the polynomial f in x_i of degree k* . Let the polynomial f have degree s in x_i . In view of the corollary to Lemma 2, the polynomial $f_i^{(k)}(x_1, \dots, x_n; x_j)$ is homogeneous in x_j of degree k for $k \leq s$ and equals zero for $k > s$. If f is a homogeneous polynomial of type $[k_1, \dots, k_i, \dots, k_n]$, then the polynomial $f_i^{(k)}(x_1, \dots, x_n; x_{n+1})$ is also homogeneous of type $[k_1, \dots, k_i - k, \dots, k_n, k]$ (for $k \leq k_i$). We note here that for any $y \in \Phi[X]$ the element $f \Delta_i^k(y)$ is obtained by substituting the element y in place of the variable x_j in the polynomial $f_i^{(k)}(x_1, \dots, x_n; x_j)$.

PROPOSITION 3. If a variety $\mathfrak{M}$ is homogeneous, then for any $f \in T(\mathfrak{M})$ and for any $y \in \Phi[X]$ the following containment is valid:

$$f \Delta(y) = \{f \Delta_i^k(y) \mid i = 1, 2, \dots; k = 0, 1, 2, \dots\} \subseteq T(\mathfrak{M}).$$

PROOF. We note first of all that if $f(x_1, \dots, x_n) \in T(\mathfrak{M})$, then also $f(y_1, \dots, y_n) \in T(\mathfrak{M})$ for any elements $y_1, \dots, y_n \in \Phi[X]$. Therefore for the proof of the containment $f \Delta_i^k(y) \in T(\mathfrak{M})$, it suffices to prove that $f_i^{(k)}(x_1, \dots, x_n; x_j) \in T(\mathfrak{M})$, where $x_j \in X \setminus \{x_1, \dots, x_n\}$. By Lemma 3 we have $T(\mathfrak{M}) \ni f(x_1, \dots, x_i + x_j, \dots, x_n) \sum_{k=0}^{n_i} f \Delta_i^k(x_j)$, where n_i is the degree of f in x_i . As was already noted, each of the polynomials $f \Delta_i^k(x_j)$ is homogeneous in x_j of degree k . In view of the homogeneity of $\mathfrak{M}$ it thus follows that $f \Delta_i^k(x_j) = f_i^{(k)}(x_1, \dots, x_n; x_j) \in T(\mathfrak{M})$, and hence also $f \Delta_i^k(y) = f_i^{(k)}(x_1, \dots, x_n; y) \in T(\mathfrak{M})$. This proves the proposition.

It is easy to see that the converse assertion is also valid: if $f \Delta(y) \subseteq T(\mathfrak{M})$ for any $f \in T(\mathfrak{M})$ and $y \in \Phi[X]$, then the variety $\mathfrak{M}$ is homogeneous. We show now that, in fact, for the homogeneity of a variety $\mathfrak{M}$, the realization of a much weaker condition is sufficient.

Let f be a nonassociative polynomial. We denote by $f \Delta$ the set of all polynomials from $\Phi[X]$ which can be obtained from f by means of consecutive partial linearizations:

$$f \Delta = \{g \in \Phi[X] \mid g = f \Delta_{i_1}^{j_1}(x_{k_1}) \cdots \Delta_{i_s}^{j_s}(x_{k_s})\}.$$

THEOREM 4. Let a variety $\mathfrak{M}$ have a system I of defining identities such that $f \Delta \subseteq T(\mathfrak{M})$ for all $f \in I$. Then $\mathfrak{M}$ is homogeneous.

PROOF. We note first of all that we can assume without loss of generality the system I consists of homogeneous polynomials. In fact, let $f = f(x_1, \dots, x_n) \in I$ and $f = f_1 + f_2$ where f_1 is homogeneous in x_1 of degree m and f_2 has degree less than m in x_1 . Further, let x_j be a generator not appearing in the listing of f . Then we have $f \Delta_1^m(x_j) = f_1 \Delta_1^m(x_j) + f_2 \Delta_1^m(x_j) = f_1(x_j, x_2, \dots, x_n) \in f \Delta$, and furthermore $f_1(x_j, x_2, \dots, x_n) \Delta_j^m(x_1) = f_1(x_1, x_2, \dots, x_n) = f_1 \in f \Delta \subseteq T(\mathfrak{M})$, i.e., $f_1, f_2 \in T(\mathfrak{M})$. We show that $f_1 \Delta \cup f_2 \Delta \subseteq T(\mathfrak{M})$. Since

$f_1 \in f \Delta$, it is clear that $f_1 \Delta \subseteq f \Delta \subseteq T(\mathfrak{M})$. Now let $g \in f_2 \Delta$. Then $g = f_2 \Delta_1 \cdots \Delta_k$, where $\Delta_i = \Delta_{t_i}^{r_i}(x_{s_i})$. We have $g = (f - f_1) \Delta_1 \cdots \Delta_k = f \Delta_1 \cdots \Delta_k - f_1 \Delta_1 \cdots \Delta_k \in T(\mathfrak{M})$. Thus we can replace the polynomial f in the system I by the polynomials f_1 and f_2 . Continuing this process, we arrive at a homogeneous system I' which satisfies the condition of the theorem. According to Lemma 1 the homogeneity of the variety $\mathfrak{M}$ is equivalent to the ideal $T(\mathfrak{M})$ having a system of homogeneous generators. By the corollary to Theorem 2 this ideal is generated by elements of the form $f(g_1, g_2, \dots, g_n)$, where $f \in I$ and $g_1, g_2, \dots, g_n \in \Phi[X]$. We decompose the polynomials $g_1, \dots, g_n$ into monomials, $g_i = \sum_{k=1}^{s_i} g_{ik}$. Then by Lemma 3 the element $h = f(g_1, \dots, g_n)$ is representable in the form

$$h = \sum_m f_m(g_{11}, \dots, g_{1s_1}, \dots, g_{n1}, \dots, g_{ns_n}),$$

where $f_m = f_m(x_{11}, \dots, x_{1s_1}, \dots, x_{n1}, \dots, x_{ns_n}) \in f \Delta$. By the homogeneity of f , all of the elements f_m are homogeneous. In addition, by assumption they are in $T(\mathfrak{M})$. But then all the elements

$$f_m(g_{11}, \dots, g_{1s_1}, \dots, g_{n1}, \dots, g_{ns_n})$$

are also homogeneous and belong to $T(\mathfrak{M})$. Consequently, the collection of these elements is the desired system of homogeneous generators for the ideal $T(\mathfrak{M})$.

This proves the theorem.

We now clarify under what sort of conditions the partial linearizations of some identity are themselves identities.

THEOREM 5. Let $f \in \Phi[X]$ be an identity of a Φ -algebra A , which is homogeneous in x_i of degree k . Suppose there are $k-1$ elements $\alpha_1, \alpha_2, \dots, \alpha_{k-1}$ in Φ such that for the determinant

$$d = \begin{vmatrix} \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{k-1} \\ \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{k-1} & \alpha_{k-1}^2 & \cdots & \alpha_{k-1}^{k-1} \end{vmatrix} = \left(\prod_{s=1}^{k-1} \alpha_s \right) \left(\prod_{i < j} (\alpha_j - \alpha_i) \right)$$

the equation $d \cdot a = 0$ implies $a = 0$ for any $a \in A$. Then all partial linearizations in x_i of the polynomial f are identities of the algebra A .

PROOF. Let $f = f(x_1, \dots, x_n)$ and $a_1, a_2, \dots, a_n, a_j$ be arbitrary elements of the algebra A . As in the proof of Theorem 3, for brevity we denote the elements $f_i^{(m)}(a_1, \dots, a_n; a_j)$ by $f_i^{(m)}(a)$. By Lemma 3 we have

$$0 = f(a_1, \dots, a_{i-1}, a_i + a_j, a_{i+1}, \dots, a_n) = \sum_{s=0}^k f_i^{(s)}(a) = \sum_{s=1}^{k-1} f_i^{(s)}(a),$$

since

$$f_i^{(0)}(a) = f(a_1, \dots, a_{i-1}, a_i, a_{i+1}, \dots, a_n) = 0$$

and

$$f_i^{(k)}(a) = f(a_1, \dots, a_{i-1}, a_j, a_{i+1}, \dots, a_n) = 0.$$

Substituting successively into the obtained equation the elements $\alpha_1 a_j$, $\alpha_2 a_j, \dots, \alpha_{k-1} a_j$ in place of a_j , we obtain

$$\begin{aligned} \alpha_1 f_i^{(1)}(a) + \alpha_1^2 f_i^{(2)}(a) + \dots + \alpha_1^{k-1} f_i^{(k-1)}(a) &= 0, \\ \alpha_2 f_i^{(1)}(a) + \alpha_2^2 f_i^{(2)}(a) + \dots + \alpha_2^{k-1} f_i^{(k-1)}(a) &= 0, \\ &\vdots \\ \alpha_{k-1} f_i^{(1)}(a) + \alpha_{k-1}^2 f_i^{(2)}(a) + \dots + \alpha_{k-1}^{k-1} f_i^{(k-1)}(a) &= 0, \end{aligned}$$

whence it follows that $d \cdot f_i^{(s)}(a) = 0$ for $1 \leq s \leq k-1$. This means $f_i^{(s)}(a) = 0$, which, since the elements $a_1, \dots, a_n, a_j \in A$ are arbitrary, proves the validity of the identity $f_i^{(s)}(x_1, x_2, \dots, x_n; x_j)$ in A . This proves the theorem.

COROLLARY. If a variety $\mathfrak{M}$ of Φ -algebras is defined by a system I of homogeneous identities of degree $\leq k$ in each of the variables and there are $k-1$ elements $\alpha_1, \alpha_2, \dots, \alpha_{k-1}$ in Φ such that for $d = (\prod_{s=1}^{k-1} \alpha_s)(\prod_{i < j} (\alpha_j - \alpha_i))$, any algebra $A \in \mathfrak{M}$, and $a \in A$ the equation $d \cdot a = 0$ implies $a = 0$, then the variety $\mathfrak{M}$ is homogeneous. In particular, every variety defined by homogeneous identities of degree ≤ 2 in each of the variables is homogeneous.

PROOF. If there can be found nonzero elements $\alpha_1, \alpha_2, \dots, \alpha_{k-1}$ in Φ , then by Theorem 5 we shall have $f \Delta \subseteq T(\mathfrak{M})$ for each $f \in I$. Hence by Theorem 4 the variety $\mathfrak{M}$ is homogeneous. If all the elements in I have degree not greater than 2 in each of the variables, then we can take the identity element 1 as α_1 and apply the first part of the corollary.

The varieties of associative, alternative, and right alternative algebras are defined by homogeneous identities of degree not greater than 2, and therefore they are homogeneous. The defining identities for the variety of Jordan algebras are also homogeneous, but one of them has degree 3 in x_1 . This is the reason that for some rings of operators Φ , for example the field $\mathbf{Z}_2$ of two elements, the variety of Jordan algebras is not homogeneous. If the equation $2x = 1$ is solvable in Φ , however, then the situation changes radically. In fact, if we set $\alpha_1 = 1$ and as α_2 take the solution of this equation (we denote it by $\frac{1}{2}$), then for $d = 1 \cdot \frac{1}{2} \cdot (1 - \frac{1}{2}) = (\frac{1}{2})^2$ the implication $da = 0 \Rightarrow a = 0$ holds, since $4da = a$. By the corollary to Theorem 5, in this case we conclude that the variety of Jordan algebras is homogeneous.

THEOREM 6. Let A be an algebra over a ring Φ which belongs to a homogeneous variety $\mathfrak{M}$ and let B be an associative-commutative Φ -algebra. Then the Φ -algebra $C = B \otimes_{\Phi} A$ also belongs to the variety $\mathfrak{M}$.

PROOF. It suffices to show that C satisfies all homogeneous identities from $T(\mathfrak{M})$. Let $f(x_1, x_2, \dots, x_n)$ be such an identity. Then for any $c_1, c_2, \dots, c_n \in C$ the element $f(c_1, c_2, \dots, c_n)$ is representable in the form of a sum of elements of the form $g(b_1 \otimes a_1, \dots, b_s \otimes a_s)$, $a_i \in A$, $b_j \in B$, where g is a homogeneous polynomial from $f \Delta \subseteq T(\mathfrak{M})$. If the polynomial $g(x_1, x_2, \dots, x_s)$ has type $[i_1, i_2, \dots, i_s]$, then

$$g(b_1 \otimes a_1, \dots, b_s \otimes a_s) = b_1^{i_1} b_2^{i_2} \cdots b_s^{i_s} \otimes g(a_1, a_2, \dots, a_s) = 0.$$

But this means that also $f(c_1, c_2, \dots, c_n) = 0$. This proves the theorem.

5. MULTILINEAR IDENTITIES. COMPLETE LINEARIZATION OF IDENTITIES

In this section we shall prove the existence of some identity in an algebra A also implies the existence of some multilinear identity in the algebra A .

We introduce the following notation: a roof $\hat{}$ over x_i in a sum $x_1 + \cdots + \hat{x}_i + \cdots + x_n$ denotes that x_i appears in the sum with zero coefficient, that is,

$$x_1 + \cdots + \hat{x}_i + \cdots + x_n = \sum_{k=1}^n x_k - x_i.$$

For example, $x_1 + \hat{x}_2 + x_3 + \hat{x}_4 + x_5 = x_1 + x_3 + x_5$.

Let $f = f(x_1, x_2, \dots, x_n)$ be some nonassociative polynomial from $\Phi[X]$ and $y_1, y_2, \dots, y_k \in X \setminus \{x_1, x_2, \dots, x_n\}$. For each $i = 1, 2, \dots, n$ we define a nonassociative polynomial fL_i^k by the formula

$$\begin{aligned} fL_i^k(x_1, \dots, x_{i-1}, y_1, y_2, \dots, y_k, x_{i+1}, \dots, x_n) \\ &= f(x_1, \dots, x_{i-1}, y_1 + y_2 + \cdots + y_k, x_{i+1}, \dots, x_n) \\ &\quad - \sum_{q=1}^k f(x_1, \dots, x_{i-1}, y_1 + \cdots + \hat{y}_q + \cdots + y_k, x_{i+1}, \dots, x_n) \\ &\quad + \sum_{1 \leq q_1 \leq q_2 \leq k} f(x_1, \dots, x_{i-1}, y_1 + \cdots + \hat{y}_{q_1} + \cdots + \hat{y}_{q_2} + \cdots \\ &\quad + y_k, x_{i+1}, \dots, x_n) \\ &\quad - \cdots + (-1)^{k-1} \sum_{q=1}^k f(x_1, \dots, x_{i-1}, y_q, x_{i+1}, \dots, x_n). \end{aligned}$$

The following is obvious.

PROPOSITION 4. Let P be a subsemigroup and Q be a subgroup of the additive group of the Φ -algebra A . Then if for a nonassociative polynomial $f = f(x_1, x_2, \dots, x_n)$ and any elements $a_1, a_2, \dots, a_n \in P$ we have $f(a_1, a_2, \dots, a_n) \in Q$, then for any $a_1, \dots, a_{i-1}, b_1, b_2, \dots, b_k, a_{i+1}, \dots, a_n \in P$ we shall also have $fL_i^k(a_1, \dots, a_{i-1}, b_1, b_2, \dots, b_k, a_{i+1}, \dots, a_n) \in Q$. In particular, if $f \in T(A)$, then also $fL_i^k \in T(A)$.

Thus the operator L_i^k transforms an identity of the algebra A into an identity again of the algebra A . In order to elucidate the mechanics of its operation, we first prove a lemma.

LEMMA 4. Let $g: A \times A \times \dots \times A \rightarrow A$ be a function on n variables, which is defined for a Φ -algebra A and is linear in each argument. Then for any $a_1, a_2, \dots, a_k \in A$, where $k \geq n$,

$$\begin{aligned} & g(a_1 + a_2 + \dots + a_k, \dots, a_1 + a_2 + \dots + a_k) \\ & - \sum_{q=1}^k g(a_1 + \dots + \hat{a}_q + \dots + a_k, \dots, a_1 + \dots + \hat{a}_q + \dots + a_k) \\ & + \sum_{1 \leq q_1 < q_2 \leq k} g(a_1 + \dots + \hat{a}_{q_1} + \dots + \hat{a}_{q_2} + \dots + a_k, \dots, a_1 + \dots \\ & + \hat{a}_{q_1} + \dots + \hat{a}_{q_2} + \dots + a_k) \\ & + \dots + (-1)^{k-1} \sum_{q=1}^k g(a_q, a_q, \dots, a_q) \\ & = \begin{cases} \sum_{(i_1 \dots i_n) \in S_n} g(a_{i_1}, a_{i_2}, \dots, a_{i_n}), & \text{if } k = n, \\ 0, & \text{if } k > n. \end{cases} \end{aligned}$$

PROOF. Let $k \geq n$. Using the linearity of the function g , we remove all $+$ signs from within the arguments of this function. Then the left side of the equality to be proved is itself represented as a linear combination of elements of the form $g(a_{j_1}, a_{j_2}, \dots, a_{j_n})$ with integer coefficients. If there are s different indices among $j_1, j_2, \dots, j_n$ and $s < k$, then the coefficient for $g(a_{j_1}, a_{j_2}, \dots, a_{j_n})$ equals the alternating sum

$$1 - \binom{k-s}{1} + \binom{k-s}{2} - \dots + (-1)^{k-s} \binom{k-s}{k-s},$$

which equals zero. We note that s does not exceed n , and k by assumption is not less than n . Therefore $s \geq k$ can occur only for $s = k = n$. In this case the coefficient for $g(a_{j_1}, a_{j_2}, \dots, a_{j_n})$ clearly equals one. This proves our formula, and consequently also the lemma.

PROPOSITION 5. For any $f, f' \in \Phi[X]$ on which the operator L_i^k is defined, $(f + f')L_i^k = fL_i^k + f'L_i^k$. If $f = f(x_1, x_2, \dots, x_m)$ is a monomial of degree n in x_i and $g = g(x_1, \dots, x_{i-1}, y_1, y_2, \dots, y_n, x_{i+1}, \dots, x_m)$ is a monomial linear in $y_1, y_2, \dots, y_n \in X \setminus \{x_1, \dots, x_m\}$ such that $f(x_1, x_2, \dots, x_m) = g(x_1, \dots, x_{i-1}, x_i, \dots, x_i, x_{i+1}, \dots, x_m)$, then

$$fL_i^k(x_1, \dots, x_{i-1}, z_1, \dots, z_k, x_{i+1}, \dots, x_m) = \begin{cases} \sum_{(i_1 \dots i_n) \in S_n} g(x_1, \dots, x_{i-1}, z_{i_1}, \dots, z_{i_n}, x_{i+1}, \dots, x_m), & \text{if } k = n, \\ 0, & \text{if } k > n. \end{cases}$$

PROOF. The linearity of the operator L_i^k follows directly from the definition. If f and g are the monomials indicated in the assumptions of the proposition, then fixing $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n$ we can consider the monomial g as a function of the n variables $y_1, y_2, \dots, y_n$. Applying Lemma 4 we obtain the formula for fL_i^k . This proves the proposition.

We illustrate this proposition by examples:

$$\begin{aligned} [x_1^2(x_2x_1)]L_1^3 &= (y_1y_2)(x_2y_3) + (y_2y_1)(x_2y_3) + (y_1y_3)(x_2y_2) + (y_3y_1)(x_2y_2) \\ &\quad + (y_2y_3)(x_2y_1) + (y_3y_2)(x_2y_1), \\ [(x_1^2x_2)x_1^5]L_1^5 &= 0, \\ [x_1^2x_2^2]L_1^2L_2^2 &= (y_1y_2)(z_1z_2) + (y_2y_1)(z_1z_2) + (y_1y_2)(z_2z_1) + (y_2y_1)(z_2z_1). \end{aligned}$$

Let $f = f(x_1, x_2, \dots, x_n)$ be a homogeneous polynomial of type $[k_1, k_2, \dots, k_n]$. Then we shall call the multilinear polynomial $fL_1^{k_1}L_2^{k_2} \dots L_n^{k_n}$ the *complete linearization of the polynomial f* .

THEOREM 7. Let a nonassociative polynomial $f = f(x_1, x_2, \dots, x_n)$ vanish for any substitution of elements from a subgroup P of the additive group of the algebra A . Then the complete linearization of any of its homogeneous components with maximal degree also vanishes on P .

PROOF. Let $f = f_1 + f_2 + \dots + f_s$ be a decomposition of the polynomial f into a sum of homogeneous components, and also let the degree of f_1 be maximal. Suppose that some variable x_j does not appear in f_1 but does appear in other homogeneous components. Then the polynomial

$$f'(x_1, x_2, \dots, x_n) = f(x_1, \dots, x_{j-1}, 0, x_{j+1}, \dots, x_n)$$

likewise vanishes on P , has f_1 as a homogeneous component of maximal degree, but f' does not contain x_j . Furthermore, if x_j appears in f_1 but does not appear in some other homogeneous component of the polynomial f' ,

then we can consider the new polynomial

$$f''(x_1, x_2, \dots, x_n) = f'(x_1, x_2, \dots, x_n) - f'(x_1, \dots, x_{j-1}, 0, x_{j+1}, \dots, x_n).$$

This polynomial also vanishes on P , but x_j now appears in all homogeneous components of the polynomial f'' , and, as before, f_1 is a homogeneous component of maximal degree in f'' .

Thus, renumbering the variables if necessary, we can assume at the outset that the homogeneous component f_1 of the polynomial f has type $[k_1, k_2, \dots, k_n]$, where $k_i \neq 0$ for $i = 1, 2, \dots, n$, and the remaining homogeneous components (f_2 for example) have type $[m_1, m_2, \dots, m_n]$, where $m_i \neq 0$ for $i = 1, 2, \dots, n$, and where the strict inequality $m_j < k_j$ holds for some j . Since $m_j < k_j$, then $f_2 L_1^{k_1} L_2^{k_2} \dots L_n^{k_n} = 0$ by Proposition 5. Consequently,

$$f_1 L_1^{k_1} L_2^{k_2} \dots L_n^{k_n} = f L_1^{k_1} L_2^{k_2} \dots L_n^{k_n}.$$

By Proposition 4 we now conclude that the complete linearization

$$f_1 L_1^{k_1} L_2^{k_2} \dots L_n^{k_n}$$

of the component f_1 vanishes on P . This proves the theorem.

COROLLARY. If an algebra A satisfies some identity, then it also satisfies some multilinear identity.

6. ADJOINING AN IDENTITY ELEMENT

Let A be some algebra over the ring Φ . We consider the ring Φ as a module over itself. The identity element 1 of the ring Φ is a generating element for this Φ -module, $\Phi = \Phi \cdot 1$. We consider the direct sum $A^\# = A \oplus \Phi \cdot 1$ of the two Φ -modules A and $\Phi \cdot 1$, and we define a multiplication on it in the following manner:

$$(a + \alpha \cdot 1)(b + \beta \cdot 1) = (ab + \alpha b + \beta a) + \alpha \beta \cdot 1,$$

where $\alpha, \beta \in \Phi$ and $a, b \in A$. We shall call the algebra $A^\#$ the *algebra obtained by the formal adjoining of an identity element to the algebra A* . It is easy to see that 1 is an identity element for the algebra $A^\#$ and A is a subalgebra in $A^\#$. We call a variety $\mathfrak{M}$ *unitally closed* if, for any algebra A from $\mathfrak{M}$, the algebra $A^\#$ also belongs to $\mathfrak{M}$.

We shall subsequently call the algebra $\Phi[X]^\#$, obtained by adjoining an identity element to the free algebra $\Phi[X]$, the *free algebra with identity element*.

PROPOSITION 6. A variety $\mathfrak{M}$ defined by set of identities I is unitally closed if and only if $I(\Phi[X]^\#) \subseteq T(\mathfrak{M})$.

PROOF. We consider the $\mathfrak{M}$ -free algebra $\Phi_{\mathfrak{M}}[X]$ and the $\mathfrak{M}$ -free algebra with identity element $\Phi_{\mathfrak{M}}[X]^{\#}$. As is easy to see, $\mathfrak{M}$ is unittally closed if and only if $\Phi_{\mathfrak{M}}[X]^{\#} \in \mathfrak{M}$. The latter condition is equivalent to $I(\Phi_{\mathfrak{M}}[X]^{\#}) = (0)$. We note now that the algebra $\Phi_{\mathfrak{M}}[X]^{\#}$ is isomorphic to the quotient algebra $\Phi[X]^{\#}/T(\mathfrak{M})$, and therefore the equality $I(\Phi_{\mathfrak{M}}[X]^{\#}) = (0)$ is equivalent to the containment $I(\Phi[X]^{\#}) \subseteq T(\mathfrak{M})$. This proves the proposition.

We denote by Δ_i^k the mapping of the algebra $\Phi[X]$ into the algebra $\Phi[X]^{\#}$, which pairs the polynomial

$$f = f(x_1, \dots, x_n)$$

from $\Phi[X]$ with the element

$$f \Delta_i^k(1) = f_i^{(k)}(x_1, \dots, x_n; 1)$$

from the algebra $\Phi[X]^{\#}$. The mapping Δ_i^k is also defined linearly on the whole algebra $\Phi[X]$. The action of the operator Δ_i^k is analogous to "partial differentiation" with respect to the variable x_i . For example,

$$\begin{aligned} [(x_1^2 x_2) x_1] \Delta_1^1 &= x_1^2 x_2 + 2(x_1 x_2) x_1, \\ [(x_1^2 x_2) x_1] \Delta_1^2 &= x_2 x_1 + 2x_1 x_2, \\ [(x_1^2 x_2) x_1] \Delta_1^3 &= x_2, \\ [(x_1^2 x_2) x_1] \Delta_1^4 &= 0. \end{aligned}$$

If the elements of the set X are not equipped with indices, then we shall replace the index i by the notation for the corresponding generator. For example,

$$[(x^2 y) x] \Delta_x^3 = y.$$

We shall denote by $f \Delta'$ the collection of all elements which can be obtained from the polynomial f by applying the operators Δ_i^k .

THEOREM 8. Let a variety $\mathfrak{M}$ have a system I of defining identities such that $f \Delta' \subseteq T(\mathfrak{M})$ for all $f \in I$. Then $\mathfrak{M}$ is unittally closed.

PROOF. In view of Proposition 6, we need to prove that $I(\Phi[X]^{\#}) \subseteq T(\mathfrak{M})$. Since $T(\mathfrak{M})$ is an ideal in $\Phi[X]^{\#}$, it suffices to prove that $f(g_1, \dots, g_n) \in T(\mathfrak{M})$ for any $f = f(x_1, \dots, x_n) \in I$ and any $g_1, \dots, g_n \in \Phi[X]^{\#}$. We have $g_i = y_i + \alpha_i \cdot 1$, where $y_i \in \Phi[X]$, $\alpha_i \in \Phi$, $i = 1, 2, \dots, n$. Now by Lemma 3

$$\begin{aligned} f(g_1, \dots, g_n) &= f(y_1 + \alpha_1 \cdot 1, \dots, y_n + \alpha_n \cdot 1) \\ &= \sum_{i_1, \dots, i_n} \alpha_1^{i_1} \cdots \alpha_n^{i_n} f \Delta_1^{i_1} \cdots \Delta_n^{i_n}(y_1, \dots, y_n). \end{aligned}$$

Since, by assumption, $f \Delta_1^{i_1} \cdots \Delta_n^{i_n} = f \Delta_1^{i_1} \cdots \Delta_n^{i_n}(x_1, \dots, x_n) \in T(\mathfrak{M})$, then also $f(g_1, \dots, g_n) \in T(\mathfrak{M})$. This proves the theorem.

We call a variety $\mathfrak{M}$ *strongly homogeneous* if $\mathfrak{M}$ is homogeneous and $f \Delta' \subseteq T(\mathfrak{M})$ for any $f \in T(\mathfrak{M})$.

PROPOSITION 7. A variety $\mathfrak{M}$ is strongly homogeneous if and only if $\mathfrak{M}$ is homogeneous and unittally closed.

PROOF. If $\mathfrak{M}$ is strongly homogeneous, then the assertion follows from Theorem 8. Conversely, let $\mathfrak{M}$ be homogeneous and unittally closed. We consider $f = f(x_1, \dots, x_n) \in T(\mathfrak{M})$. By Proposition 3 we have $f \Delta_i^k(x_j) = f_i^{(k)}(x_1, \dots, x_n; x_j) \in T(\mathfrak{M})$, where $x_j \in X \setminus \{x_1, \dots, x_n\}$. Since the set of identities $T(\mathfrak{M})$ defines the variety $\mathfrak{M}$, then by Proposition 6 we obtain $f \Delta_i^k = f_i^{(k)}(x_1, \dots, x_n; 1) \in T(\mathfrak{M})$. This proves the proposition.

From Theorems 4 and 8 we now obtain

COROLLARY 1. Let the variety $\mathfrak{M}$ have a system I of defining identities such that $f \Delta \cup f \Delta' \subseteq T(\mathfrak{M})$ for all $f \in I$. Then $\mathfrak{M}$ is strongly homogeneous.

In particular, we have

COROLLARY 2. The varieties of associative, alternative, and right alternative algebras are strongly homogeneous. If there is an element $\frac{1}{2}$ in Φ , then the variety of Jordan Φ -algebras is also strongly homogeneous.

PROOF. As we have already seen in Section 4, all of these varieties are homogeneous. We show that the variety Jord of Jordan algebras is strongly homogeneous. This variety is defined by the identities $f_1 = x_1 x_2 - x_2 x_1$ and $f_2 = (x_1^2 x_2) x_1 - x_1^2 (x_2 x_1)$. It is easy to see that $f_i \Delta_i^k = 0$ in all cases except when $i = 2, k = l = 1$. In that case

$$\begin{aligned} f_2 \Delta_1^1 &= 2(x_1 x_2) x_1 + x_1^2 x_2 - 2x_1(x_2 x_1) - x_1^2 x_2 \\ &= 2(x_1 x_2 - x_2 x_1) x_1 + 2(x_2 x_1) x_1 - 2x_1(x_2 x_1) \\ &= 2f_1(x_1, x_2) x_1 + 2f_1(x_2 x_1, x_1) \in T(\text{Jord}). \end{aligned}$$

Thus $f_1 \Delta' \cup f_2 \Delta' \subseteq T(\text{Jord})$, and the variety Jord is strongly homogeneous. The proofs for the other varieties are analogous. This proves the corollary.

We now prove a property for the free algebras of strongly homogeneous varieties.

PROPOSITION 8. Let $\mathfrak{M}$ be a strongly homogeneous variety. Then for any $f \in \Phi_{\mathfrak{M}}[X]$ and for any $x \in X$ each of the equalities $f \cdot x = 0, x \cdot f = 0$ implies $f = 0$.

PROOF. We denote by $d_x(f)$ the degree of the polynomial f in the variable x , and by R_z the operator of right multiplication by the element z , $yR_z = y \cdot z$. We shall show by induction on $d_x(f)$ that for any natural number m the equality $fR_x^m = 0$ implies $f = 0$. Because of the strong homogeneity of the variety, for $d_x(f) = 0$ we have $0 = (fR_x^m)\Delta_x^m = f$, which gives us the basis for the induction. Let $d_x(f) = n > 0$. Using again the strong homogeneity of the variety $\mathfrak{M}$ and Lemma 2, we obtain

$$0 = (fR_x^m)\Delta_x^m = \sum_{i=0}^m \binom{m}{i} (f\Delta_x^i)R_x^i = f + \sum_{i=1}^m \binom{m}{i} (f\Delta_x^i)R_x^i,$$

whence $f = f_1R_x$, where $d_x(f_1) < n$. We now have

$$0 = fR_x^m = f_1R_x^{m+1}.$$

The induction assumption gives us $f_1 = 0$ and $f = 0$.

In particular, we have proved that if $f \cdot x = 0$, then $f = 0$. It is proved analogously that the equality $x \cdot f = 0$ implies $f = 0$.

This proves the proposition.

An element a of an algebra A is called a *right (left) annihilator* of A if $A \cdot a = (0)$ ($a \cdot A = (0)$, respectively).

COROLLARY. A free algebra of a strongly homogeneous variety does not contain nonzero right and left annihilators.

In particular, free alternative, right alternative, and Jordan algebras do not contain nonzero annihilators.

In conclusion we prove a proposition which sometimes facilitates the proof of identities in strongly homogeneous varieties. Its assertion is often called the *Koecher principle*.

PROPOSITION 9. Let $\mathfrak{M}$ be a strongly homogeneous variety and $f = f(x_1, \dots, x_n)$ be some homogeneous nonassociative polynomial. Suppose that for any algebra A from $\mathfrak{M}$ which is finitely generated as a Φ -module, the polynomial f vanishes for any substitution of elements of the form $1 + x$, where the ideal generated in A by the element x is nilpotent. Then f is an identity of the variety $\mathfrak{M}$.

PROOF. It suffices to establish that $f(x_1, \dots, x_n) = 0$ in the $\mathfrak{M}$ -free algebra $F = \Phi_{\mathfrak{M}}[x_1, \dots, x_n]$. Let the degree of the polynomial f equal m . We consider the quotient algebra $G = F/F^{m+1}$ and the elements $y_1, \dots, y_n$ in G that are images of the elements $x_1, \dots, x_n$ under the canonical homomorphism of F onto G . Adjoining an identity element to this algebra, we obtain the algebra $G^{\#} \in \mathfrak{M}$. The elements $y_1, \dots, y_n$ generate nilpotent ideals

in this algebra. Therefore

$$f(1 + y_1, \dots, 1 + y_n) = 0.$$

But then, by strong homogeneity, in the algebra $F^\#$ we have the relation

$$f(1 + x_1, \dots, 1 + x_n) = 0.$$

The polynomial $f(x_1, \dots, x_n)$ is a homogeneous component of the polynomial $f(1 + x_1, \dots, 1 + x_n)$, and therefore equals zero. This proves the proposition.

In all the classical varieties elements of the form indicated in the conditions of the proposition are invertible, so that in these varieties it suffices to prove identities only for invertible elements.

CHAPTER 2

Composition Algebras

One of the most important properties of the complex numbers is expressed by the identity

$$|zz'| = |z| \cdot |z'|.$$

If we denote $z = a_1 + a_2i$, $z' = b_1 + b_2i$, then this identity is rewritten in the form

$$(a_1^2 + a_2^2)(b_1^2 + b_2^2) = (a_1b_1 - a_2b_2)^2 + (a_1b_2 + a_2b_1)^2.$$

Put somewhat loosely, this formula can be read as the product of a sum of two squares times the sum of two squares is again the sum of two squares.

The question that naturally arises is: Do there exist analogous identities for a greater number of squares? How are all such identities described?

Examples of identities for the sum of four squares were already known to Euler and Lagrange. In 1818 Degen[†] found the first example of an identity for the sum of eight squares, which, unfortunately, then went unnoticed. In 1843 Hamilton noted that the existence of an identity for the sum of n squares is equivalent to the existence of a division algebra of a definite form with dimension n over the field of real numbers. For example, the algebra of

[†] Degen C. F., *Mem. Acad. Sci. St. Petersburg* 8, (1818), 207–219.

complex numbers is such an algebra in the case $n = 2$. Hamilton constructed a four-dimensional division algebra which gave Euler's identity for the sum of four squares. In 1845 Cayley constructed an eight-dimensional division algebra giving an identity for the sum of eight squares. An analogous algebra was constructed independently by Graves in 1844. It was not clarified until much later whether or not analogous identities exist for other values of n . Only in 1898 did Hurwitz prove the identities of interest to us are possible only for $n = 1, 2, 4, 8$.

The algebras constructed by Hamilton and Cayley, now known, respectively, as the algebra of quaternions and algebra of Cayley numbers, also proved to be very interesting from many other points of views. Thus in 1878 Frobenius proved that the algebra of quaternions and the field of complex numbers are the only finite-dimensional associative division algebras with dimension greater than 1 over the field of real numbers. Later the algebra of Cayley numbers found an important place in the theory of alternative and Jordan algebras.

In connection with the Hurwitz theorem arises a more general question. What happens if, instead of the sum of squares, some nondegenerate quadratic form over an arbitrary field is considered? For which quadratic forms do results analogous to the Hurwitz theorem hold? This question, known as the Hurwitz problem, was solved comparatively recently in works of Albert, Kaplansky, and Jacobson. It turns out that in this case the problem also reduces to the description of a certain class of algebras—the so-called “composition algebras.” Composition algebras are a natural generalization of the algebras of complex numbers, quaternions, and Cayley numbers, and they play an important role in the theory of alternative and Jordan algebras. This chapter is devoted to the study of these algebras.

1. DEFINITION AND SIMPLE PROPERTIES OF COMPOSITION ALGEBRAS

Let F be an arbitrary field and A be a vector space of nonzero dimension over F . A mapping $f: A \times A \rightarrow F$ is called a *bilinear form* if for any $x, x', y, y' \in A$ and $\alpha \in F$

- (1) $f(x + x', y) = f(x, y) + f(x', y)$;
- (2) $f(x, y + y') = f(x, y) + f(x, y')$;
- (3) $f(\alpha x, y) = f(x, \alpha y) = \alpha f(x, y)$.

A bilinear form f is called *symmetric* if $f(x, y) = f(y, x)$ for all $x, y \in A$. A symmetric bilinear form f is called *nondegenerate* if, from $f(a, x) = 0$ for all $x \in A$, it follows that $a = 0$.

A mapping $n: A \rightarrow F$ is called a *quadratic form* if

- (1) $n(\lambda x) = \lambda^2 n(x)$, where $x \in A$, $\lambda \in F$;
- (2) the function $f(x, y) = n(x + y) - n(x) - n(y)$ is a bilinear form on A .

A quadratic form $n(x)$ is called *strictly nondegenerate* if the symmetric bilinear form $f(x, y)$ which corresponds to it is nondegenerate, and it is called *nondegenerate* if from $n(a) = f(a, x) = 0$ for all $x \in A$ it follows that $a = 0$. Every strictly nondegenerate quadratic form is nondegenerate, and the converse is true if the characteristic of the field F is different from 2. A form $n(x)$ is said to *admit composition* if there exists a bilinear binary operation (composition) xy in A such that

$$n(x)n(y) = n(xy). \quad (1)$$

The *Hurwitz problem* consists of the determination of nondegenerate quadratic forms admitting composition. If $A = E^n$ is n -dimensional Euclidean space and $n(x) = (x, x)$ is the scalar square of the vector x , then we obtain the classical formulation of the Hurwitz problem. In this case, equality (1) written in an orthogonal basis gives the desired identity for the sum of squares.

As an example of a quadratic form admitting composition and different from the sum of squares, the following form from two variables can be used:

$$n(x) = n(x_1, x_2) = x_1^2 + kx_1x_2 + tx_2^2, \quad k, t \in F.$$

Actually, it is easy to verify that

$$n(x_1, x_2) \cdot n(y_1, y_2) = n(x_1y_1 - tx_2y_2, x_1y_2 + x_2y_1 + kx_2y_2).$$

If the form $n(x)$ admits composition, then the vector space A becomes an algebra with respect to the operations of addition, scalar multiplication, and the multiplication xy . In addition, the multiplication is connected with the quadratic form $n(x)$ by equality (1). Conversely, if a quadratic form $n(x)$ connected with the multiplication by equality (1) is defined on some algebra A , then the form $n(x)$ admits composition. Consequently, the Hurwitz problem is equivalent to the problem of describing the algebras on which there is defined a nondegenerate quadratic form $n(x)$ satisfying (1).

As we shall see later, if the form $n(x)$ is strictly nondegenerate and admits composition, then as in the Hurwitz theorem the dimension of the corresponding algebra A can only equal 1, 2, 4, 8. At the same time, over each imperfect field of characteristic 2 there exists an infinite number of nondegenerate quadratic forms which are not strictly nondegenerate and do admit composition. In addition, the dimension of the corresponding algebra A can be an infinite number of different values and can even be infinite. In this connection, in order to protect ourselves from "pathological" cases in

characteristic 2 and at the same time not impose a restriction on the characteristic, we shall subsequently consider only strictly nondegenerate quadratic forms. Nondegenerate quadratic forms admitting composition are considered in the exercises for Section 2.

An algebra A over a field F with quadratic form $n(x)$ is called a *composition algebra* if

- (1) $n(xy) = n(x)n(y)$;
- (2) the form $n(x)$ is strictly nondegenerate;
- (3) there is an identity element 1 in A .

PROPOSITION 1. Let $n(x)$ be a strictly nondegenerate quadratic form on a finite-dimensional vector space A . Then if $n(x)$ admits some kind of composition xy on A , $n(x)$ admits a composition $x \cdot y$ with respect to which A is a composition algebra.

PROOF. By assumption, the algebra A with multiplication xy and the form $n(x)$ satisfy conditions (1) and (2). By condition (2) there exists an element $a \in A$ for which $n(a) \neq 0$. Let $u = a^2/n(a)$. Then $n(u) = 1$, and consequently $n(xu) = n(ux) = n(x)$. Hence it follows that $f(xu, yu) = f(ux, uy) = f(x, y)$. By condition (2) this means that the linear mappings $R_u: x \rightarrow xu$ and $L_u: x \rightarrow ux$ are nonsingular. In view of the finite dimensionality of A , the mappings R_u and L_u are invertible. We now define a new multiplication on A by setting $x \cdot y = (xR_u^{-1})(yL_u^{-1})$. Then $u^2 \cdot x = (u^2R_u^{-1})(xL_u^{-1}) = u(xL_u^{-1}) = x$. Analogously, $x \cdot u^2 = x$. Consequently, u^2 is an identity element with respect to the new multiplication. Furthermore,

$$n(x \cdot y) = n((xR_u^{-1})(yL_u^{-1})) = n(xR_u^{-1})n(yL_u^{-1}) = n(x)n(y).$$

Thus we can replace the composition xy by $x \cdot y$ and obtain a composition algebra. This proves the proposition.

Thus, for strictly nondegenerate quadratic forms defined on finite-dimensional spaces, the Hurwitz problem is equivalent to the problem of describing finite-dimensional composition algebras.

We shall describe all composition algebras without any sort of restriction on the dimension.

We shall need the following definition.

An algebra R is called *alternative* if $x^2y = x(xy)$, $yx^2 = (yx)x$ for all x, y from R .

LEMMA 1. Let A be a composition algebra. Then A is alternative and each element from the algebra A satisfies a quadratic equation with coefficients from F (i.e., the algebra A is quadratic over F).

PROOF. We shall identify the field F with the subalgebra $F \cdot 1$ of the algebra A . Substituting $y + w$ in place of y in (1), we obtain $n(x)n(y + w) = n(xy + xw)$. Subtracting identity (1) from this equality, and also the identity obtained from (1) by replacing y by w , we obtain

$$n(x)f(y, w) = f(xy, xw). \quad (2)$$

Performing the same procedure with x , we obtain

$$f(x, z)f(y, w) = f(xy, zw) + f(xw, zy). \quad (3)$$

We now substitute $z = 1$, $y = xu$ in (3)[†]:

$$f(x, 1)f(xu, w) = f(x \cdot xu, w) + f(xw, xu). \quad (4)$$

Since by (2) $f(xw, xu) = n(x)f(w, u)$, then (4) can be rewritten in the form

$$f(x \cdot xu, w) + n(x)f(w, u) - f(x, 1)f(xu, w) = 0,$$

which by the bilinearity and symmetry of the form f is equivalent to the identity

$$f(x \cdot xu + n(x)y - f(x, 1)xu, w) = 0.$$

Since the form $f(x, y)$ is nondegenerate and w is arbitrary, it follows from this that

$$x \cdot xu + n(x)u - f(x, 1)xu = 0 \quad (5)$$

for any x, y from A . Now putting $u = 1$ in (5), we obtain

$$x^2 - f(x, 1)x + n(x) = 0, \quad (6)$$

which proves the second half of the lemma. It just remains to prove the algebra A is alternative.

Multiplying (6) on the right by u and comparing with (5), we obtain $x^2u = x(xu)$. It is proved analogously that $ux^2 = (ux)x$. Consequently the algebra A is alternative, which proves the lemma.

An endomorphism ρ of a vector space A is called an *involution* of the algebra A if $\rho(\rho(a)) = a$ and $\rho(ab) = \rho(b)\rho(a)$ for all $a, b \in A$.

LEMMA 2. The mapping $a \rightarrow \bar{a} = f(1, a) - a$ is an involution of the algebra A which leaves elements of the field F fixed. Also, the elements $t(a) = a + \bar{a}$ and $n(a) = a\bar{a}$ lie in F for all a from A . Moreover, a satisfies the equality

$$a^2 - t(a)a + n(a) = 0.$$

[†] Here and subsequently, in order to avoid a profusion of parentheses we shall sometimes use dots in place of some of them; for example, $xy \cdot zt = (xy)(zt)$.

PROOF. We shall verify, in turn, each of the properties for an involution:

(a) $\overline{a+b} = \bar{a} + \bar{b}$. We have $\overline{a+b} = f(1, a+b) - (a+b) = f(1, a) - a + f(1, b) - b = \bar{a} + \bar{b}$.

(b) Let $\lambda \in F$. Then $\overline{\lambda a} = f(1, \lambda a) - \lambda a = \lambda(f(1, a) - a) = \lambda \bar{a}$.

(c) $\bar{\bar{a}} = a$. We have $\bar{\bar{a}} = f(1, \bar{a}) - \bar{a} = f(1, f(1, a) - a) - f(1, a) + a = f(1, 1)f(1, a) - 2f(1, a) + a$. Furthermore, as is easy to see, $n(1) = 1$. Therefore, $f(1, 1) = n(2) - n(1) - n(1) = 2$. Consequently, $\bar{\bar{a}} = a$.

(d) $\overline{ab} = \bar{b}\bar{a}$. From identity (6), substituting in turn $a+b$, a , and b in place of x , and then subtracting from the first identity obtained the other two, we have

$$ab + ba - f(1, a)b - f(1, b)a + f(a, b) = 0. \quad (7)$$

Furthermore, from (3) we obtain for $x = y = 1$, $z = a$, and $w = b$,

$$f(1, a)f(1, b) = f(1, ab) + f(a, b). \quad (8)$$

Substituting this in (7), we obtain

$$ab + ba - f(1, a)b - f(1, b)a + f(1, a)f(1, b) - f(1, ab) = 0, \quad (9)$$

whence

$$\begin{aligned} (f(1, a) - a)(f(1, b) - b) &= ab - f(1, a)b - f(1, b)a + f(1, a)f(1, b) \\ &= f(1, ab) - ba. \end{aligned}$$

From (8) it follows that $f(1, ab) = f(1, ba)$. Consequently $\overline{ab} = \bar{b}\bar{a}$, which is what was to be proved.

Furthermore, if $\lambda \in F$, then $\overline{\lambda} = f(1, \lambda) - \lambda = \lambda f(1, 1) - \lambda = \lambda$. Finally, $a + \bar{a} = f(1, a) \in F$, and by (6) $a\bar{a} = f(1, a)a - a^2 = n(a) \in F$. Since $t(a) = f(1, a)$, the last assertion of the lemma also follows from (6). This proves the lemma.

LEMMA 3. Let A be an algebra over a field F with identity element 1 and with involution $x \rightarrow \bar{x}$, where for every $x \in A$ the elements $x + \bar{x}$ and $x\bar{x}$ belong to F . Then if the quadratic form $n(x) = x\bar{x}$ is strictly nondegenerate on A , the algebra A is either simple or isomorphic to the direct sum $F \oplus F$ with involution $(\bar{a}, \bar{b}) = (b, a)$.

PROOF. We prove first that A is simple as an algebra with involution. Let $I \neq A$ be an ideal of the algebra A and $\bar{I} = I$. Since $I \cap F = (0)$, we have $a + \bar{a} = 0$, $a\bar{a} = 0$ for any $a \in I$. Furthermore, for any elements $a \in I$, $x \in A$ we have $a\bar{x} \in I$, whence by the preceding $f(a, x) = a\bar{x} + x\bar{a} = a\bar{x} + (\bar{a}\bar{x}) = 0$. In view of the nondegeneracy of the form f , it follows from this that $I = (0)$.

Now let T be an arbitrary ideal of the algebra A , $T \neq (0)$ and $T \neq A$. We set $I = T + \bar{T}$. Then $\bar{I} = I$ and $I \neq (0)$, whence by what was proved above

$I = A$. Furthermore, we set $J = T \cap \bar{T}$. Then $\bar{J} = J$ and $J \neq A$, and again by what was proved above $J = (0)$. Consequently, the algebra A is the direct sum of the ideals T and $\bar{T}$. It remains to prove that the ideal T is one-dimensional over the field F . Let $t, s \in T$ with $t \neq 0$, and $\lambda = t + \bar{t}$, $\mu = s + \bar{s}$ so that $\lambda, \mu \in F$. We note that $\lambda \neq 0$, since otherwise $t = -\bar{t} \in T \cap \bar{T} = (0)$. Furthermore, we have $T \cdot \bar{T} + \bar{T} \subseteq T \cap \bar{T} = (0)$, whence $\lambda s = (t + \bar{t})s = ts$, $\mu t = t(s + \bar{s}) = ts$. Consequently, $\lambda s = \mu t$ and $s = (\lambda^{-1}\mu)t$, that is, the one-dimensionality of the ideal T is proved.

Thus, if the algebra A is not simple, then it is the direct sum of two copies of the field F . This proves the lemma.

We see that the problem of describing composition algebras brings us in a natural manner to the study of simple alternative algebras. In Section 3 we show that, in turn, with the exception of some algebras of characteristic 2, each simple quadratic alternative algebra is a composition algebra. In Chapter 7 we shall obtain a definitive description of simple alternative algebras that are not associative.

2. THE CAYLEY-DICKSON PROCESS. GENERALIZED THEOREM OF HURWITZ

Let A be an algebra over a field F with identity element 1 and with an involution $a \rightarrow \bar{a}$, where $a + \bar{a}, a\bar{a} \in F$ for all $a \in A$. By means of the so-called *Cayley-Dickson process* we shall construct a new algebra with involution which contains A as a subalgebra. In addition, if the dimension of the algebra A equals m , then the dimension of the new algebra will equal $2m$.

We fix $0 \neq \alpha \in F$, and we denote by (A, α) the collection of all ordered pairs (a_1, a_2) , $a_i \in A$, with operations of componentwise addition and scalar multiplication and the multiplication

$$(a_1, a_2)(a_3, a_4) = (a_1a_3 + \alpha a_4\bar{a}_2, \bar{a}_1a_4 + a_3a_2).$$

As is easy to see, (A, α) is an algebra over F . The element $(1, 0)$ is an identity element of the algebra (A, α) . The set $A' = \{(a, 0) | a \in A\}$ is a subalgebra of the algebra (A, α) which is isomorphic to the algebra A . Let $v = (0, 1)$. Then $v^2 = \alpha(1, 0)$, and (A, α) is a direct sum of the vector spaces A' and vA' . If we identify A' with A , the elements of the algebra (A, α) are represented in the form $x = a_1 + va_2$, where the $a_i \in A$ are uniquely determined by the element x and the multiplication in (A, α) is given by

$$(a_1 + va_2)(a_3 + va_4) = (a_1a_3 + \alpha a_4\bar{a}_2) + v(\bar{a}_1a_4 + a_3a_2).$$

For an arbitrary element $x = a_1 + va_2 \in (A, \alpha)$, we set $\bar{x} = \bar{a}_1 - va_2$.

LEMMA 4. The mapping $x \rightarrow \bar{x}$ is an involution of the algebra (A, α) . In addition, $x + \bar{x}, x\bar{x} \in F$ for all $x \in (A, \alpha)$. If the quadratic form $n(a) = a\bar{a}$ is strictly nondegenerate on A , then the quadratic form $n(x) = x\bar{x}$ is strictly nondegenerate on (A, α) .

PROOF. It is clear that the mapping $x \rightarrow \bar{x}$ is linear and that $\bar{\bar{x}} = x$ for any $x \in (A, \alpha)$. Let $x = a_1 + va_2, y = a_3 + va_4$, where $a_i \in A$. Then we have

$$\bar{y}\bar{x} = (\bar{a}_3 - va_4)(\bar{a}_1 - va_2) = (\bar{a}_3\bar{a}_1 + \alpha a_2\bar{a}_4) - v(a_3a_2 + \bar{a}_1a_4).$$

On the other hand,

$$\overline{xy} = (\bar{a}_3\bar{a}_1 + \alpha a_2\bar{a}_4) - v(\bar{a}_1a_4 + a_3a_2) = \bar{y}\bar{x}.$$

Consequently the mapping $x \rightarrow \bar{x}$ is an involution of the algebra (A, α) . Furthermore, $x + \bar{x} = a_1 + \bar{a}_1 \in F$ and $x\bar{x} = a_1\bar{a}_1 - a_2\bar{a}_2 \in F$.

Now let the quadratic form $n(a)$ be strictly nondegenerate on A . The bilinear form $f(x, y) = x\bar{y} + y\bar{x} = (a_1\bar{a}_3 + a_3\bar{a}_1) - \alpha(a_2\bar{a}_4 + a_4\bar{a}_2)$, where $x = a_1 + va_2, y = a_3 + va_4$, and $a_i \in A$, corresponds to the quadratic form $n(x) = x\bar{x}$. Let $f(x, y) = 0$ for all $y \in (A, \alpha)$. Setting $y = a_3 \in A$, we obtain $a_1\bar{a}_3 + a_3\bar{a}_1 = 0$ for all $a_3 \in A$. Hence, in view of the strict nondegeneracy of the form $n(a) = a\bar{a}$, it follows $a_1 = 0$. Since $\alpha \neq 0$, analogously we obtain $a_2 = 0$, whence $x = 0$. This means the quadratic form $n(x) = x\bar{x}$ is strictly nondegenerate, which proves the lemma.

Now let A be a composition algebra with quadratic form $n(a)$. By Lemma 2 there exists an involution $a \rightarrow \bar{a}$ on A such that $n(a) = a\bar{a}$ and $t(a) = a + \bar{a} \in F$ for all $a \in A$. This means we can apply the Cayley-Dickson process to A . We next find the conditions under which the obtained algebra (A, α) with quadratic form $n(x)$ is also a composition algebra.

LEMMA 5. If A is a composition algebra, then (A, α) is a composition algebra if and only if the algebra A is associative.

PROOF. It was shown above that the algebra (A, α) has an identity element. In addition, by Lemma 4 the quadratic form $n(x) = x\bar{x}$ is strictly nondegenerate on (A, α) . This means (A, α) is a composition algebra if and only if the form $n(x)$ admits composition. We note that if $x = a_1 + va_2$, then $n(x) = n(a_1) - \alpha n(a_2)$. Let $y = a_3 + va_4$. We then have

$$\begin{aligned} n(xy) - n(x)n(y) &= n(a_1a_3 + \alpha a_4\bar{a}_2) - \alpha n(\bar{a}_1a_4 + a_3a_2) \\ &\quad - [n(a_1) - \alpha n(a_2)][n(a_3) - \alpha n(a_4)] \\ &= n(a_1a_3) + \alpha^2 n(a_4\bar{a}_2) + \alpha f(a_1a_3, a_4\bar{a}_2) \\ &\quad - \alpha n(\bar{a}_1a_4) - \alpha n(a_3a_2) - \alpha f(a_1\bar{a}_4, a_3a_2) \\ &\quad - n(a_1)n(a_3) + \alpha n(a_1)n(a_4) + \alpha n(a_2)n(a_3) \\ &\quad - \alpha^2 n(a_2)n(a_4). \end{aligned}$$

Since $n(ab) = n(a)n(b)$ and $n(\bar{a}) = n(a)$ for any $a, b \in A$, we obtain

$$n(xy) - n(x)n(y) = \alpha f(a_1 a_3, a_4 \bar{a}_2) - \alpha f(\bar{a}_1 a_4, a_3 a_2).$$

In view of (3) we have

$$\begin{aligned} f(a_1 a_3, a_4 \bar{a}_2) &= f(a_1 a_3 \cdot 1, a_4 \bar{a}_2) = -f(a_1 a_3 \cdot \bar{a}_2, a_4) + f(a_1 a_3, a_4) f(1, \bar{a}_2) \\ &= f(a_1 a_3 \cdot (f(1, \bar{a}_2) - \bar{a}_2), a_4) = f(a_1 a_3 \cdot a_2, a_4), \end{aligned}$$

and analogously

$$f(\bar{a}_1 a_4, a_3 a_2) = f(a_4, a_1 \cdot a_3 a_2).$$

Consequently

$$n(xy) - n(x)n(y) = \alpha f(a_1 a_3 \cdot a_2 - a_1 \cdot a_3 a_2, a_4).$$

Since $\alpha \neq 0$ and the bilinear form f is nondegenerate, it follows from this that the quadratic form $n(x)$ admits composition if and only if the algebra A is associative. This proves the lemma.

We can now cite the following examples of composition algebras.

I. F is a field of characteristic $\neq 2$, and $n(\alpha) = \alpha^2$ for all $\alpha \in F$. We note that in the case of characteristic 2, the field F is not a composition algebra. In fact, in that case the bilinear form $f(x, y) = (x + y)^2 - x^2 - y^2 \equiv 0$ on F , that is, the quadratic form $n(x) = x^2$ is not strictly nondegenerate. It is possible to change the definition of a composition algebra so that a field of characteristic 2 is a composition algebra, but there then appears an entire supplement of composition algebras of characteristic 2, which can even be infinite-dimensional (see Exercise 6).

II. $K(\mu) = F + Fv$, where $\overline{v_1^2} = v_1 + \mu$ and $4\mu + 1 \neq 0$, the field F is arbitrary, the involution is $\alpha + \beta v_1 = (\alpha + \beta) - \beta v_1$, and the quadratic form is $n(a) = a\bar{a}$. If the polynomial $x^2 - x - \mu$ is irreducible in $F[x]$, then the algebra $K(\mu)$ is a field (a separable quadratic extension of the field F); otherwise $K(\mu) = F \oplus F$. If the characteristic of the field F does not equal 2, then the element $v = v_1 - \frac{1}{2}$ satisfies the equality $v^2 = \alpha$ where $\alpha = \frac{1}{4}(4\mu + 1) \neq 0$. In this case $K(\mu) = (F, \alpha)$. Conversely, if F is a field of characteristic $\neq 2$, then the algebra (F, α) is an algebra of type II. In fact, setting $v_1 = v + \frac{1}{2}$ we obtain $\overline{v_1^2} = v_1 + (\alpha - \frac{1}{4})$, and moreover $4(\alpha - \frac{1}{4}) + 1 = 4\alpha \neq 0$. In addition, $\gamma + \beta v_1 = \gamma + \beta(v + \frac{1}{2}) = \gamma - \beta v + \frac{1}{2}\beta = (\gamma + \beta) - \beta v_1$. We note here that if F is a field of characteristic 2, then the condition $4\mu + 1 \neq 0$ is automatically satisfied.

III. $\mathcal{Q}(\mu, \beta) = (K(\mu), \beta)$ with $\beta \neq 0$ is the algebra of generalized quaternions. As is easy to see, the algebra $\mathcal{Q}(\mu, \beta)$ is associative but not commutative.

IV. $C(\mu, \beta, \gamma) = (Q(\mu, \beta), \gamma)$ with $\gamma \neq 0$ is the *Cayley–Dickson algebra*. It is easy to verify (see Exercise 2) that the Cayley–Dickson algebra $C(\mu, \beta, \gamma)$ is not associative, and therefore by Lemma 5 it is impossible to continue further our inductive process for the construction of composition algebras.

We shall now prove that all the composition algebras are accounted for by our indicated examples.

Let A be a composition algebra and $n(x)$ be the corresponding quadratic form defined on A . By Lemma 2 there is an involution $a \rightarrow \bar{a}$ on the algebra A such that $n(a) = a\bar{a}$ and $a + \bar{a} \in F$ for all $a \in A$. In addition, $\bar{\bar{a}} = a$ for any $a \in A$. Let $f(x, y)$ be the bilinear form associated with the quadratic form $n(x)$. Then $f(x, y) = x\bar{y} + y\bar{x}$. If B is some subspace of the space A , we shall denote by $B^\perp$ the orthogonal complement of the subspace B with respect to the form $f(x, y)$, $B^\perp = \{a \in A \mid f(a, B) = 0\}$.

We shall need the following:

LEMMA 6. Let B be a subalgebra of a composition algebra A which contains the identity element 1 of the algebra A . Then $B^\perp B + BB^\perp \subseteq B^\perp$, and for every $a, b \in B$ and $v \in B^\perp$ the following relations are valid:

$$\bar{v} = -v, \quad av = v\bar{a}, \quad (10)$$

$$a(vb) = v(\bar{a}b), \quad (vb)a = v(ab), \quad (11)$$

$$(va)(vb) = -n(v)b\bar{a}. \quad (12)$$

PROOF. From the equalities $f(v, 1) = 0$, $f(a, v) = 0$ we obtain $v + \bar{v} = 0$, $a\bar{v} + v\bar{a} = 0$, whence follows (10). Furthermore, by (3) we have

$$f(a, vb) = f(a - 1, vb) = -f(ab, v) + f(a, v)f(1, b) = 0,$$

and analogously

$$f(a, bv) = 0.$$

Since the elements $a, b \in B$ and $v \in B^\perp$ are arbitrary, this means that $BB^\perp + B^\perp B \subseteq B^\perp$. Furthermore, by Lemma 1 the identity $x(xy) = x^2y$ is satisfied in the algebra A . Hence it follows easily that A also satisfies the relation

$$x(\bar{x}y) = (x\bar{x})y = n(x)y. \quad (13)$$

Linearizing this relation in x (see Chapter 1), we obtain

$$x(\bar{z}y) + z(\bar{x}y) = f(x, z)y. \quad (14)$$

If we set here $x = a$, $y = b$, $z = v$, we obtain

$$a(\bar{v}b) = -v(\bar{a}b),$$

from which follows the first of equalities (11). Acting on both sides of this equality with the involution, we obtain the second of equalities (11). Finally, by identity (14) and in view of (2), (13), and (11), we have

$$\begin{aligned}(va)(vb) &= -\bar{v}(\bar{v}\bar{a} \cdot b) + f(va, \bar{v})b = v(\bar{a}\bar{v} \cdot b) - f(va, v)b \\ &= -v(\bar{a}v \cdot b) - n(v)f(a, 1)b = -v(va \cdot b) - n(v)f(a, 1)b \\ &= v(\bar{v} \cdot ba) - n(v)f(a, 1)b = n(v)(ba - f(a, 1)b) = -n(v)b\bar{a}.\end{aligned}$$

This proves the lemma.

We can now prove

THEOREM 1. Let A be a composition algebra. Then A is isomorphic to one of the above-mentioned algebras of types I–IV.

PROOF. By the term “subalgebra” we shall mean a subalgebra containing the identity element 1. Since $a + \bar{a} \in F$, every such subalgebra is invariant under the involution.

Let B be a finite-dimensional subalgebra of the algebra A on which the restriction of the form $f(x, y)$ is nondegenerate. As is well known (e.g., see A. I. Mal'cev, “Foundations of Linear Algebra,” p. 284, Nauka, 1975.), A then decomposes into a direct sum of subspaces $A = B + B^\perp$. Moreover, the restriction of the form $f(x, y)$ to $B^\perp$ is also nondegenerate. We assume that $B \neq A$. Then we can find a $v \in B^\perp$ such that $n(v) = -\alpha \neq 0$. From (2) we obtain $f(va, vb) = n(v)f(a, b) = -\alpha f(a, b)$. Since f is nondegenerate on B , it follows from this that the mapping $x \rightarrow vx$ of the subspace B onto vB is one-to-one. Consequently, B and vB have the same dimension. In addition, the relation obtained shows that the subspace vB is nondegenerate with respect to $f(x, y)$. This means the subspace $B_1 = B + vB$, which by Lemma 6 is the orthogonal sum of two nondegenerate subspaces, is also nondegenerate. Relations (11) and (12) show that $B_1 = B + vB$ is the subalgebra of the algebra A obtained from B by means of the Cayley–Dickson process, $B_1 = (B, \alpha)$. Since $\bar{v} = -v$ and $\overline{a + vb} = \bar{a} - \bar{b}v = \bar{a} - vb$, the involution induced on B_1 by the involution on A coincides with the involution obtained in the Cayley–Dickson process. Finally, the subalgebra B_1 is nondegenerate with respect to $f(x, y)$ and satisfies the same conditions as B . Therefore we can repeat the same process with the algebra B_1 .

We now return to the algebra A and consider separately two cases.

(1) F is a field of characteristic $\neq 2$. In this case the subalgebra F is nondegenerate with respect to $f(x, y)$, and therefore we can set $B = F$. If $F \neq A$, then there is a subalgebra B_1 of type II contained in A . If $B_1 \neq A$, then there is a subalgebra B_2 of type III contained in A . Finally, if $B_2 \neq A$, then A contains a subalgebra B_3 of type IV. With this the process must

terminate, since otherwise by Lemma 5 there would be in the algebra A a subalgebra B_4 which is not a composition algebra, and that is impossible. Consequently $A = B_3$.

(2) F is a field of characteristic 2. As we have seen, in this case $A \neq F$. We shall show there exists $a \in A \setminus F$ for which $t(a) \neq 0$. In fact, let $t(x) = f(1, x) = 0$ for every $x \in A \setminus F$. Since $t(\alpha) = 2\alpha = 0$ for every $\alpha \in F$, then in general $t(x) = f(1, x) = 0$ for all $x \in A$, which contradicts the nondegeneracy of the form f . Let $t(a) \neq 0$ with $a \notin F$, and set $s = a/t(a)$. Then $t(s) = 1$, that is, $s^2 = s + \mu$ where $\mu \in F$. As is easy to see, the subalgebra $F + Fs$ is nondegenerate with respect to $f(x, y)$. Furthermore, $\overline{\alpha + \beta s} = \alpha + \beta - \beta s$, so that the subalgebra $F + Fs$ is an algebra of type II. Setting $B = F + Fs$, we again obtain that the algebra A is one of the algebras of type II, III, or IV.

This proves the theorem.

COROLLARY. A strictly nondegenerate quadratic form $n(x)$ defined on a vector space V over a field F admits composition if and only if $\dim_F V = 1, 2, 4, 8$, and with respect to some basis for the space V the form $n(x)$ corresponds to one of the forms

- (1) $n(x) = x^2$ ($\text{char } F \neq 2$);
- (2) $n(x) = x_1^2 + x_1x_2 - \mu x_2^2$;
- (3) $n(x) = x_1^2 + x_1x_2 - \mu x_2^2 - \beta x_3^2 - \beta x_3x_4 + \beta \mu x_4^2$;
- (4) $n(x) = x_1^2 + x_1x_2 - \mu x_2^2 - \beta x_3^2 - \beta x_3x_4 + \beta \mu x_4^2$
 $- \gamma x_5^2 - \gamma x_5x_6 + \gamma \mu x_6^2 + \beta \gamma x_7^2 + \beta \gamma x_7x_8 - \beta \gamma \mu x_8^2$,

where $\mu, \beta, \gamma \in F$, $(4\mu + 1)\beta\gamma \neq 0$. If the characteristic of the field F does not equal 2, then it is possible to choose a basis for V in which the form $n(x)$ has one of the forms

- (1) $n(x) = x^2$;
- (2) $n(x) = x_1^2 - \alpha x_2^2$;
- (3) $n(x) = x_1^2 - \alpha x_2^2 - \beta x_3^2 + \alpha \beta x_4^2$;
- (4) $n(x) = x_1^2 - \alpha x_2^2 - \beta x_3^2 + \alpha \beta x_4^2 - \gamma x_5^2 + \alpha \gamma x_6^2 + \beta \gamma x_7^2 - \alpha \beta \gamma x_8^2$,

where $\alpha, \beta, \gamma \in F$, $\alpha\beta\gamma \neq 0$.

Exercises

1. Let $A = F + Fs$, where $s^2 = s + \alpha$, $\alpha \in F$, $4\alpha + 1 \neq 0$. Prove there exists a unique nontrivial involution on A , and also A is a composition algebra with respect to this involution (the algebra $K(\mu)$ of type II).

2. Let A be an algebra over a field F with identity element 1 and with involution $a \rightarrow \bar{a}$ such that $a + \bar{a}$, $\bar{a}\bar{a} \in F$ for every $a \in A$. Also, let (A, α) be the algebra obtained from A by means of the Cayley-Dickson process.

Prove that

(a) the algebra (A, α) is associative if and only if A is associative and commutative;

(b) the algebra (A, α) is alternative if and only if the algebra A is associative.

3. Prove that in a Cayley–Dickson algebra $C(\mu, \beta, \gamma)$ over a field F of characteristic $\neq 2$ it is possible to choose a basis $e_0 = 1, e_1, e_2, \dots, e_7$ with the following multiplication table ($e_0 = 1$ is the identity element of the algebra $C(\mu, \beta, \gamma)$):

	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	α	e_3	αe_2	e_5	αe_4	$-e_7$	$-\alpha e_6$
e_2	$-e_3$	β	$-\beta e_1$	e_6	e_7	βe_4	βe_5
e_3	$-\alpha e_2$	βe_1	$-\alpha\beta$	e_7	αe_6	$-\beta e_5$	$-\alpha\beta e_4$
e_4	$-e_5$	$-e_6$	$-e_7$	γ	$-\gamma e_1$	$-\gamma e_2$	$-\gamma e_3$
e_5	$-\alpha e_4$	$-e_7$	$-\alpha e_6$	γe_1	$-\alpha\gamma$	γe_3	$\alpha\gamma e_2$
e_6	e_7	$-\beta e_4$	βe_5	γe_2	$-\gamma e_3$	$-\beta\gamma$	$-\beta\gamma e_1$
e_7	αe_6	$-\beta e_6$	$\alpha\beta e_4$	γe_3	$-\alpha\gamma e_2$	$\beta\gamma e_1$	$\alpha\beta\gamma$

where $\alpha = \frac{1}{4}(4\mu + 1) \neq 0$. If $\alpha = \beta = \gamma = -1$ and F is the field of real numbers, then we obtain the classical *algebra of Cayley numbers*, which resolves the Hurwitz problem for the sum of eight squares.

4. Prove that over the field of real numbers the quadratic form $n(x) = x_1^2 + x_2^2 + x_3^2 - x_4^2$ does not admit composition.

5. Determine up to equivalence all nondegenerate quadratic forms over the field of real numbers which admit composition. (There exist seven such forms in all.)

6. Prove that if in the definition of a composition algebra the requirement of strict nondegeneracy of the form $n(x)$ is changed to nondegeneracy, then besides the algebras of types I–IV there appear the following composition algebras: a field F of characteristic 2 and arbitrary purely inseparable quadratic extensions of the field F for which the dimension either equals 2^t or is infinite. *Hint.* If $f(x, y) \equiv 0$, then the mapping $x \rightarrow n(x)$ is a ring homomorphism with zero kernel.

3. SIMPLE QUADRATIC ALTERNATIVE ALGEBRAS

In this section we shall prove that with the exception of some algebras of characteristic 2 every simple quadratic alternative algebra is a composition algebra.

We need some simple results on alternative algebras.

Let A be some algebra over an arbitrary ring of operators Φ and $x, y, z \in A$. We denote by $(x, y, z) = (xy)z - x(yz)$ the *associator* of the elements x, y, z ; by $[x, y] = xy - yx$ the *commutator* of elements x, y ; and by $x \circ y = xy + yx$ the *Jordan product* of the elements x, y . With this notation, the identities defining the variety of alternative algebras are written in the form

$$(x, x, y) = 0, \quad (x, y, y) = 0.$$

The first of these identities is called the *left alternative identity*, and the second is the *right alternative identity*.

Linearizing the left and right alternative identities, we obtain the identities

$$(x, z, y) + (z, x, y) = 0, \quad (x, y, z) + (x, z, y) = 0,$$

from which it follows that in an alternative algebra, the associator is a skew-symmetric function of its arguments. In particular, every alternative algebra satisfies the identity

$$(x, y, x) = 0.$$

Algebras satisfying this identity are called *flexible*, and this identity is itself called the flexible identity. For example, every commutative or anticommutative algebra is flexible. In view of flexibility, in an alternative algebra we can (and shall) subsequently write the product xyx without indicating an arrangement of parentheses.

LEMMA 7. In every alternative algebra A the following identities are valid:

$$\begin{aligned} x(yz) &= [(xy)x]y, & \text{the right Moufang identity,} \\ (yz)x &= y[z(yx)], & \text{the left Moufang identity,} \\ (xy)(zx) &= x(yz)x, & \text{the middle Moufang identity.} \end{aligned}$$

PROOF. We shall first prove that the algebra A satisfies the identity

$$(x^2, y, x) = 0. \tag{15}$$

By the flexible and left alternative identities we have

$$(x^2y)x = [x(xy)x] = x(xyx) = x^2(yx),$$

that is, (15) is proved. Now by (15)

$$xy^3 = (xy^2)y - (x, y^2, y) = (xy^2)y = [(xy)y]y.$$

In view of the corollary to Theorem 1.5, the variety of alternative algebras is homogeneous, and therefore by Proposition 1.3 the algebra A satisfies the

identity

$$\begin{aligned}
 0 &= \{xy^3 - [(xy)y]y\} \Delta_y^1(z) \\
 &= x(yzy) + x(y^2 \circ z) - [(xy)z]y - (xy^2)z - (xz)y^2 \\
 &= x(yzy) - [(xy)z]y - (x, y^2, z) - (x, z, y^2) \\
 &= x(yzy) - [(xy)z]y.
 \end{aligned}$$

The left Moufang identity is proved analogously. Finally,

$$\begin{aligned}
 (xy)(zx) - x(yz)x &= -(xy, z, x) + (x, y, z)x \\
 &= (z, xy, x) + (z, x, y)x \\
 &= [z(xy)]x - z(xy)x + [(zx)y]x - [z(xy)]x = 0.
 \end{aligned}$$

This proves the lemma.

COROLLARY. Every alternative algebra satisfies the identities

$$(x, xy, z) = (x, y, z)x, \quad (16)$$

$$(x, yx, z) = x(x, y, z), \quad (16')$$

$$(x^2, y, z) = (x, x \circ y, z), \quad (17)$$

$$(x^2, y, z) = x \circ (x, y, z). \quad (17')$$

PROOF. Identities (16) and (16') are equivalent to the right and left Moufang identities, respectively. In fact,

$$\begin{aligned}
 (z, xy, x) + (z, x, y)x &= -z(xy)x + [(zx)y]x, \\
 (x, yx, z) + x(y, x, z) &= (xyx)z - x[y(xz)].
 \end{aligned}$$

Furthermore, from (15) we obtain

$$0 = (x^2, y, x) \Delta_x^1(z) = (x \circ z, y, x) + (x^2, y, z),$$

whence follows (17). Finally, (17') follows from (17), (16), and (16'). This proves the corollary.

We shall subsequently also call identities (16)–(17) the *Moufang identities*.

THEOREM 2 (Artin). In an alternative algebra A any two elements generate an associative subalgebra.

PROOF. It clearly suffices to show that if $u_i = u_i(x_1, x_2)$ for $i = 1, 2, 3$, are arbitrary nonassociative words from the generators x_1, x_2 , then for any $a, b \in A$ the equality $(\bar{u}_1, \bar{u}_2, \bar{u}_3) = 0$ is valid, where $\bar{u}_i = u_i(a, b)$. We shall prove this assertion by induction on the number $n = d(u_1) + d(u_2) + d(u_3)$,

where $d(u_i)$ is the length of the word u_i . If $n = 3$, then everything follows from the left and right alternative and flexible identities. Now let $n > 3$ and for any nonassociative words $v_1(x_1, x_2)$, $v_2(x_1, x_2)$, $v_3(x_1, x_2)$ such that $d(v_1) + d(v_2) + d(v_3) < n$ let the equality $(\bar{v}_1, \bar{v}_2, \bar{v}_3) = 0$ hold. Then by the induction assumption the associator $(\bar{u}_1, \bar{u}_2, \bar{u}_3)$ does not depend on the way the parentheses are arranged in the words u_i , and therefore it is possible to assume that each u_i has parentheses arranged from the right, that is,

$$u_i = \{ \cdots [(x_1^{(i)} x_2^{(i)}) x_3^{(i)}] \cdots \} x_{k_i}^{(i)},$$

where $x_j^{(i)} \in \{x_1, x_2\}$ and $k_i = d(u_i)$. In this case some two of the words u_1, u_2, u_3 terminate in the same generator. For example, let $u_1 = (v_1)x_1$, $u_2 = (v_2)x_1$. It is easy to see that if v_1 or v_2 is missing, then by the induction assumption and the Moufang identities $(\bar{u}_1, \bar{u}_2, \bar{u}_3) = 0$. Consequently, it is possible to assume that $d(v_1) \geq 1$ and $d(v_2) \geq 1$. Then by the linearized Moufang identity we have

$$\begin{aligned} (\bar{u}_1, \bar{u}_2, \bar{u}_3) &= (\bar{v}_1 a, \bar{v}_2 a, \bar{u}_3) \\ &= -(\bar{v}_1 \bar{u}_3, \bar{v}_2 a, a) + a(\bar{v}_1, \bar{v}_2 a, \bar{u}_3) + \bar{u}_3(\bar{v}_1, \bar{v}_2 a, a) \\ &= -(\bar{v}_1 \bar{u}_3, \bar{v}_2 a, a) = -a(\bar{v}_1 \bar{u}_3, \bar{v}_2, a) = 0. \end{aligned}$$

This proves the theorem.

An algebra A is called a *power-associative algebra* if each element of the algebra A generates an associative subalgebra.

COROLLARY. Every alternative algebra is power associative.

We note that it also follows from Artin's theorem that an alternative algebra can be defined as an algebra in which any two elements generate an associative subalgebra.

We recall that an algebra A over a field F with identity element 1 is called *quadratic* over F if each element x from A satisfies an equality of the form

$$x^2 - t(x)x + n(x) = 0; \quad t(x), n(x) \in F \quad (18)$$

(in addition, we identify the field F with the subalgebra $F \cdot 1$ of the algebra A). We shall call the elements $t(x)$ and $n(x)$ the *trace* and *norm*, respectively, of the element x . By definition we set $t(\alpha) = 2\alpha$, $n(\alpha) = \alpha^2$ for $\alpha \in F$. Then the trace $t(x)$ and norm $n(x)$ are uniquely defined for each $x \in A$.

THEOREM 3. Let A be a quadratic algebra over a field F which contains at least three elements. Then the trace $t(x)$ is a linear function, and the norm $n(x)$ is a quadratic form on A . If, in addition, the algebra A is alternative, then the quadratic form $n(x)$ satisfies relation (1).

PROOF. Let $\alpha, \beta \in F$ with $\alpha \neq \beta, \alpha\beta \neq 0$. From (18) for any $x, y \in A$ we have

$$0 = (\alpha x + \beta y)^2 - t(\alpha x + \beta y)(\alpha x + \beta y) + n(\alpha x + \beta y) - \alpha\beta[(x + y)^2 - t(x + y)(x + y) + n(x + y)],$$

and furthermore,

$$\begin{aligned} 0 &= [\alpha^2 t(x) - \alpha t(\alpha x + \beta y) - \alpha\beta t(x) + \alpha\beta t(x + y)]x \\ &\quad + [\beta^2 t(y) - \beta t(\alpha x + \beta y) - \alpha\beta t(y) + \alpha\beta t(x + y)]y \\ &\quad + [-\alpha^2 n(x) - \beta^2 n(y) + n(\alpha x + \beta y) + \alpha\beta n(x) + \alpha\beta n(y) - \alpha\beta n(x + y)]. \end{aligned}$$

Now if $1, x, y$ are linearly independent over F , then we have

$$\begin{aligned} \alpha[\alpha t(x) - t(\alpha x + \beta y) - \beta t(x) + \beta t(x + y)] &= 0, \\ \beta[\beta t(y) - t(\alpha x + \beta y) - \alpha t(y) + \alpha t(x + y)] &= 0, \end{aligned}$$

whence

$$(\alpha - \beta)[t(x) + t(y) - t(x + y)] = 0$$

and

$$t(x + y) = t(x) + t(y). \quad (19)$$

Now let the elements $1, x, y$ be linearly dependent over F and, for example, $y = \alpha x + \beta$. By direct calculation we obtain from (18) $t(y) = \alpha t(x) + 2\beta$ and $t(x + y) = t((1 + \alpha)x + \beta) = (1 + \alpha)t(x) + 2\beta = t(x) + t(y)$. Thus equality (19) is proved for all x and y .

Furthermore, for any $\lambda \in F$ we have

$$\lambda^2 x^2 - \lambda t(x)\lambda x + \lambda^2 n(x) = 0,$$

whence $t(\lambda x) = \lambda t(x)$ and $n(\lambda x) = \lambda^2 n(x)$. By the same token the linearity of the trace $t(x)$ is established. For the proof that the norm $n(x)$ is a quadratic form on A , it remains for us to show that the function $f(x, y) = n(x + y) - n(x) - n(y)$ is a bilinear form on A . From the equality

$$(x + y)^2 - t(x + y)(x + y) + n(x + y) = 0,$$

in view of (19) we obtain

$$f(x, y) = t(x)y + t(y)x - x \circ y. \quad (20)$$

Hence the bilinearity of the function $f(x, y)$ follows from the linearity of the trace. At the same time the first part of Theorem 3 is proved.

Now let A be an alternative algebra. We shall prove equality (1). By Artin's theorem the subalgebra generated by elements x and y is associative.

Multiplying equality (20) on the right by xy , we obtain

$$f(x, y)xy = t(x)yxy + t(y)x^2y - (xy)^2 - yx^2y.$$

By (18) we have

$$\begin{aligned} yx^2y &= y(t(x)x - n(x))y \\ &= t(x)yxy - n(x)t(y)y + n(x)n(y), \\ t(y)x^2y &= t(y)t(x)xy - t(y)n(x)y. \end{aligned}$$

Substituting these two equalities into the previous one, we obtain

$$(xy)^2 - (-f(x, y) + t(y)t(x))xy + n(x)n(y) = 0,$$

whence follows (1) if $xy \notin F$. Now let $xy = \alpha \in F$. Since $n(\lambda x) = \lambda^2 n(x) = n(\lambda)n(x)$ for any $\lambda \in F$, it suffices for us to consider the case when $x \notin F$, $y \notin F$. Multiplying equality (18) on the right by y^2 , we obtain

$$\alpha^2 - \alpha t(x)y + n(x)y^2 = 0.$$

Let $n(x) = 0$. If in this case $\alpha \neq 0$, then we have $\alpha = t(x)y$, whence $y \in F$, which is not so. This means $\alpha = 0$ and $n(xy) = 0 = n(x)n(y)$. If $n(x) \neq 0$, then we obtain

$$\frac{\alpha^2}{n(x)} - \frac{\alpha t(x)}{n(x)}y + y^2 = 0,$$

whence in view of the fact that $y \notin F$ it follows $n(y) = \alpha^2/n(x)$, and $n(x)n(y) = \alpha^2 = n(xy)$. Equality (1) is established, and this proves the theorem.

THEOREM 4. Let A be a simple quadratic alternative algebra over a field F containing at least three elements. Then either A is a composition algebra or A is some field of characteristic 2.

PROOF. By Theorem 3 there is defined on the algebra A a quadratic form $n(x)$ connected with the multiplication in the algebra A by equality (1). Let $f(x, y)$ be the bilinear form which corresponds to the quadratic form $n(x)$. We consider the set $W = \{a \in A \mid f(a, A) = 0\}$, the kernel of the form $f(x, y)$, and we show that W is an ideal in A . It is clear that W is a subspace of the vector space A . As in the proof of Lemma 1, linearizing relation (1) we obtain that relation (3) is valid for the algebra A . Let $a \in W$ and $x, y \in A$. Then from (3) we obtain

$$f(ax, y) = f(ax, y \cdot 1) = -f(a, yx) + f(a, y)f(x, 1) = 0,$$

that is, W is a right ideal of the algebra A . It is proved analogously that W is a left ideal in A . In view of the simplicity of the algebra A , either $W = (0)$

or $W = A$. In the first case the form $n(x)$ is strictly nondegenerate, and A is a composition algebra. Now let $W = A$. In this case $f(x, y) \equiv 0$ on A , so that $n(x + y) = n(x) + n(y)$ for any $x, y \in A$. Consequently the mapping $x \rightarrow n(x)$ is a homomorphism of the ring A into the ring F . As is easy to see, the kernel of this homomorphism is an ideal of the algebra A . Since the algebra A is simple and $n(1) = 1 \neq 0$, this kernel equals zero, that is, the ring A is isomorphic to some subring of the ring F . Furthermore, from the equality $0 = f(1, 1) = 2$ it follows that in this case the field F has characteristic 2. It is also clear that the ring A is a subfield of the field F .

This proves the theorem.

Exercises

1. (Generalized theorem of Artin) Let A be an alternative algebra and $a, b, c \in A$. Prove that if $(a, b, c) = 0$, then the subalgebra generated by the elements a, b, c is associative.

In Exercises 2–4 D is a finite-dimensional alternative algebra without divisors of zero over the field of real numbers $\mathbf{R}$.

2. Prove that D is a division algebra, that is, for any $a, b \in D$ with $a \neq 0$ there are solutions to the equations

$$ax = b, \quad ya = b.$$

3. Prove that there is an identity element 1 in D .

4. Prove that the algebra D is quadratic over $\mathbf{R}$. *Hint.* Each element of the algebra D generates an associative subalgebra. Therefore the concept of a minimal polynomial for an element of a finite-dimensional algebra makes sense in D .

5. (Generalized theorem of Frobenius) Let D be a finite-dimensional alternative algebra without divisors of zero over the field of real number $\mathbf{R}$. Then D is isomorphic to one of the following algebras over $\mathbf{R}$: (1) $\mathbf{R}$; (2) the field of complex numbers $\mathbf{K}$; (3) the division ring of quaternions $\mathbf{Q}$; (4) the algebra of Cayley numbers $\mathbf{C}$.

6. Let A be an algebra with identity element 1 over a field F , and let K be some extension of the field F . Prove that if the algebra $A_K = K \otimes_F A$ is quadratic over K , then the algebra A is quadratic over F .

4. FURTHER PROPERTIES OF COMPOSITION ALGEBRAS

In this section we shall prove some properties of composition algebras which we need later. We recall that in every composition algebra A , in addition to the quadratic form $n(x)$, there are also defined the bilinear form $f(x, y)$ connected with $n(x)$ and the linear trace form $t(x) = f(1, x)$. Further-